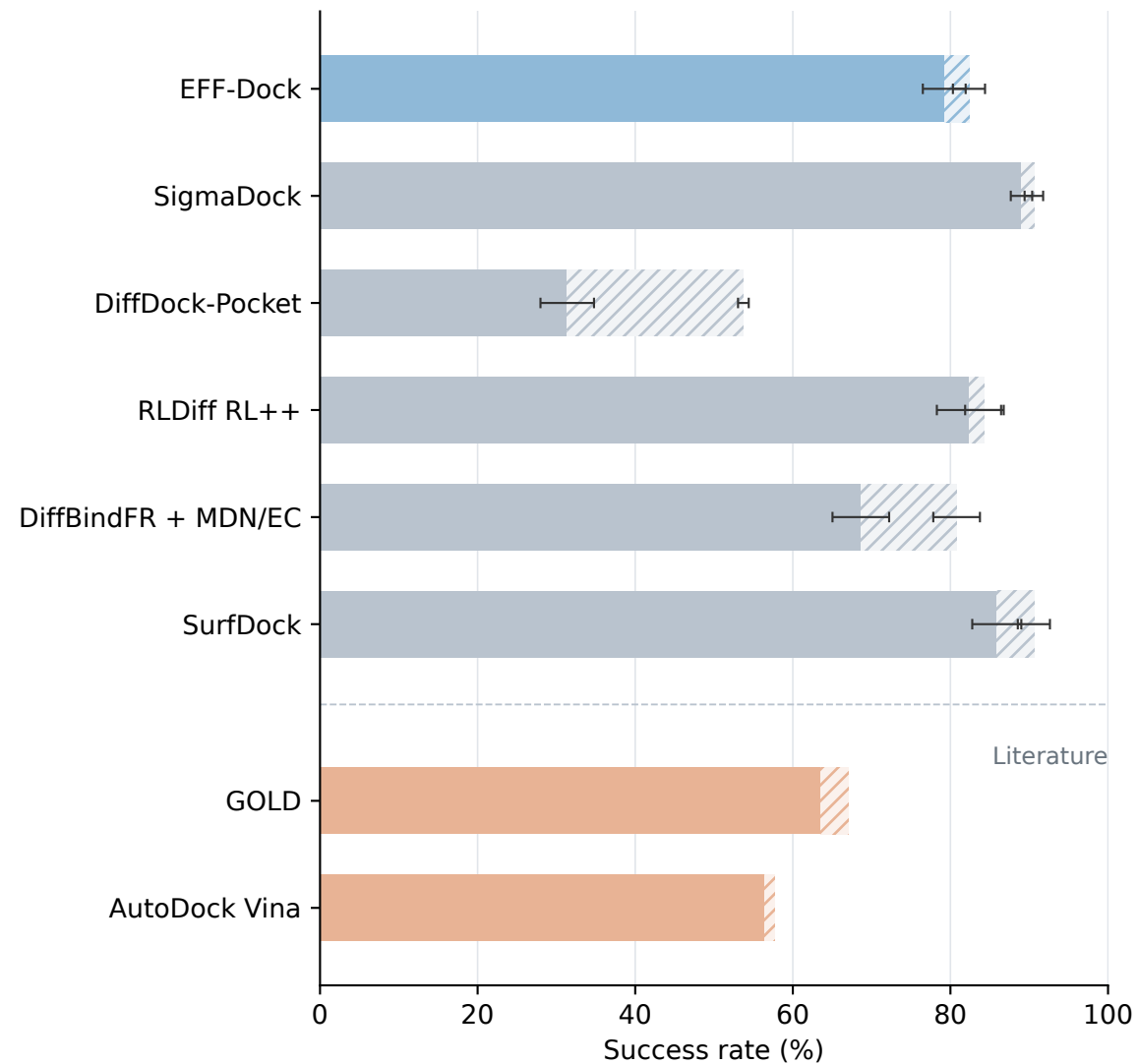
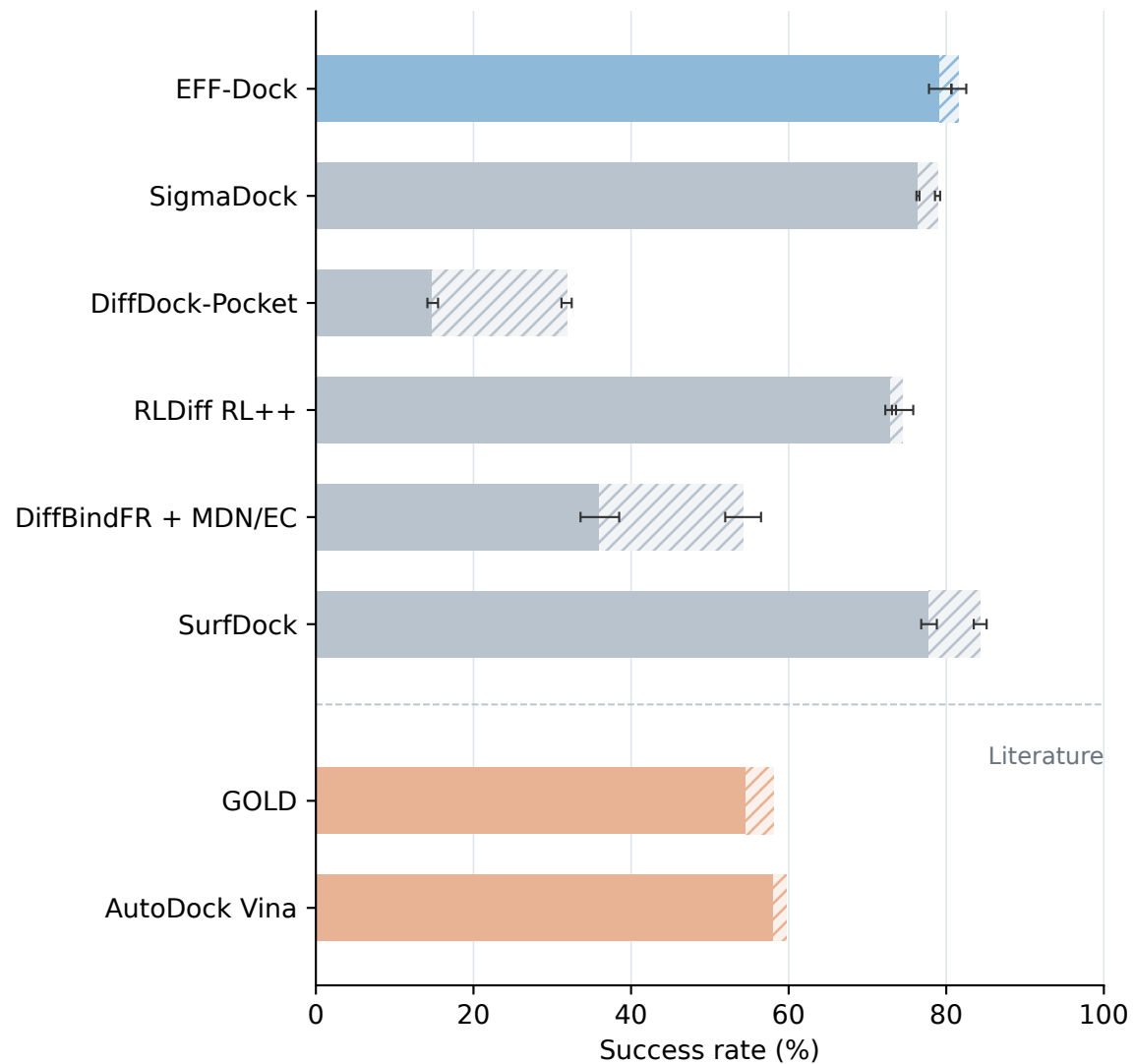
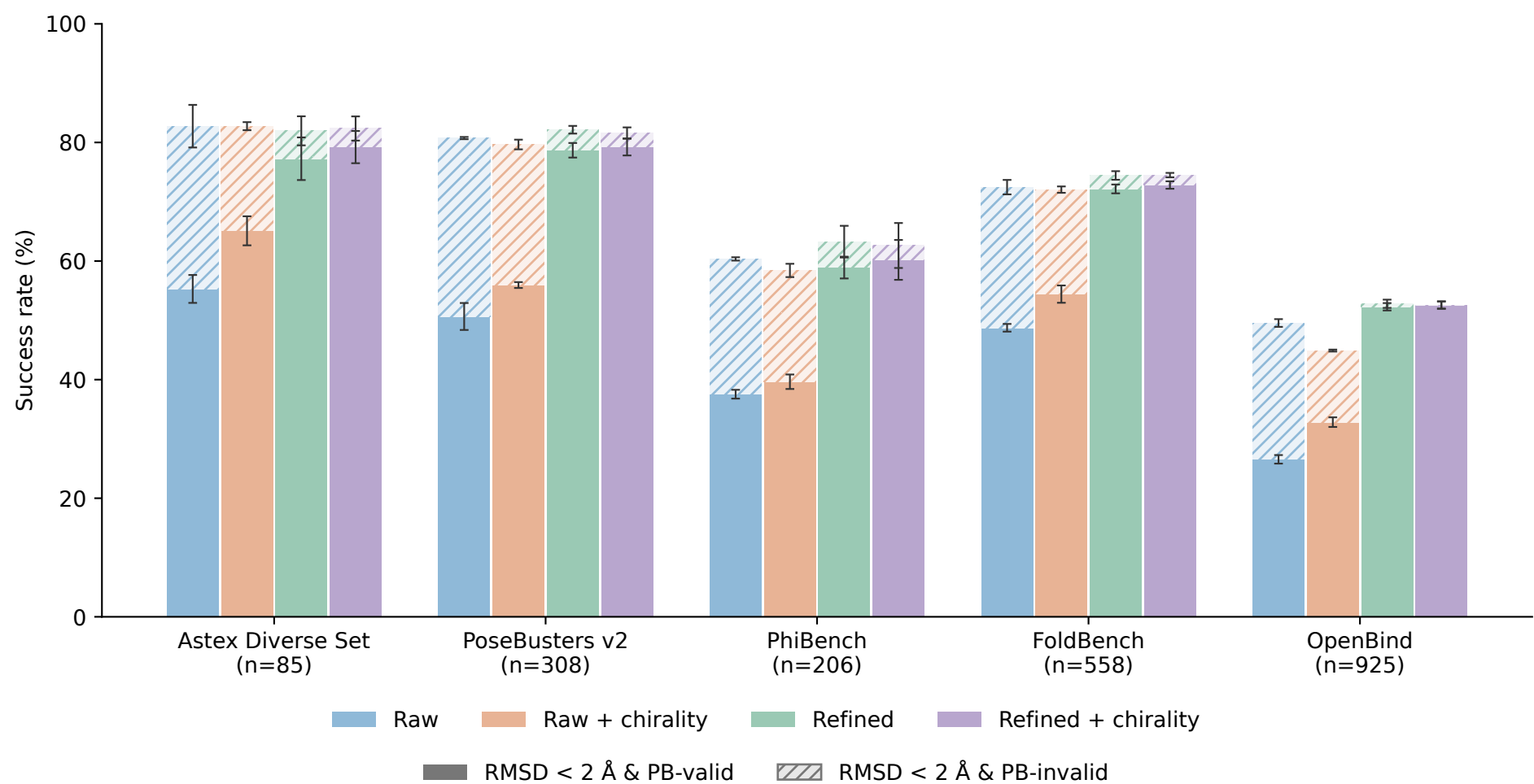
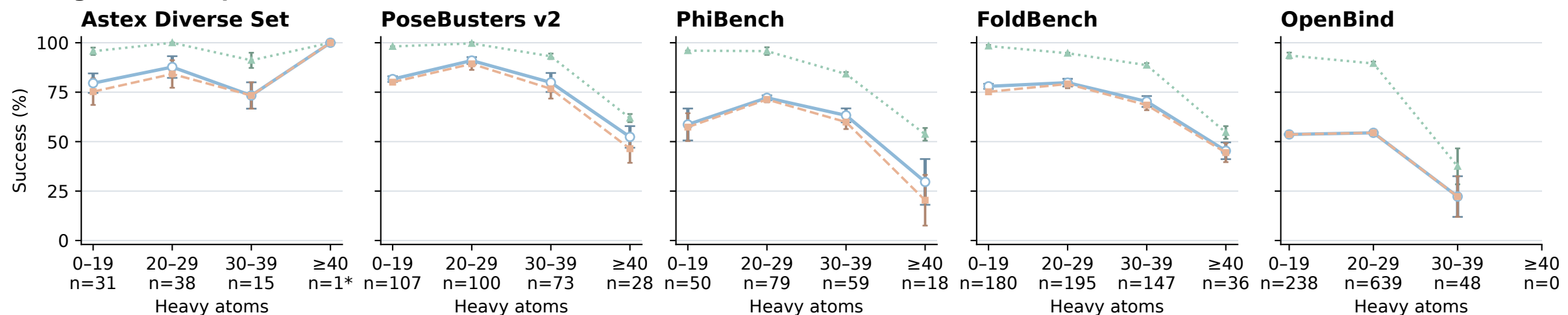


A Astex Diverse Set**B PoseBusters v2**

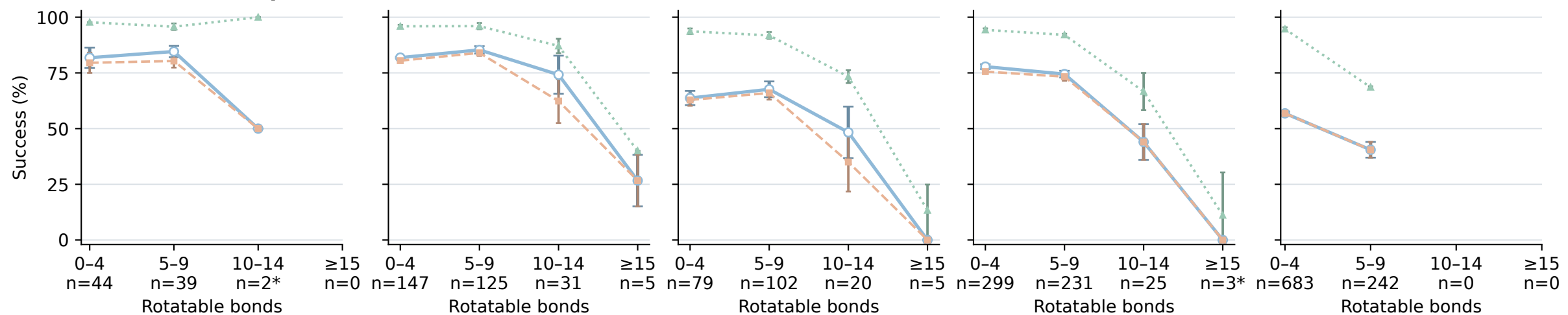
■ RMSD < 2 Å & PB-valid ▨ RMSD < 2 Å & PB-invalid



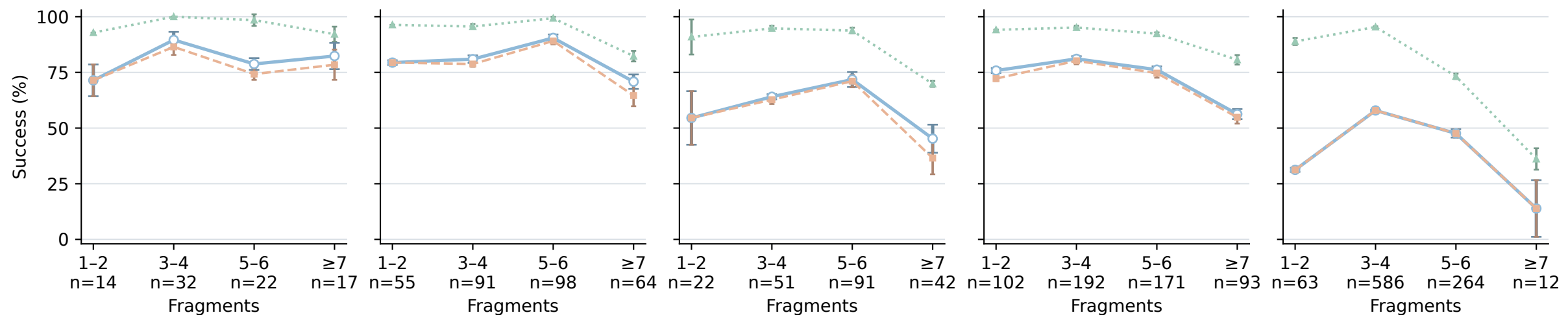
A Ligand size dependence



B Rotatable bond dependence

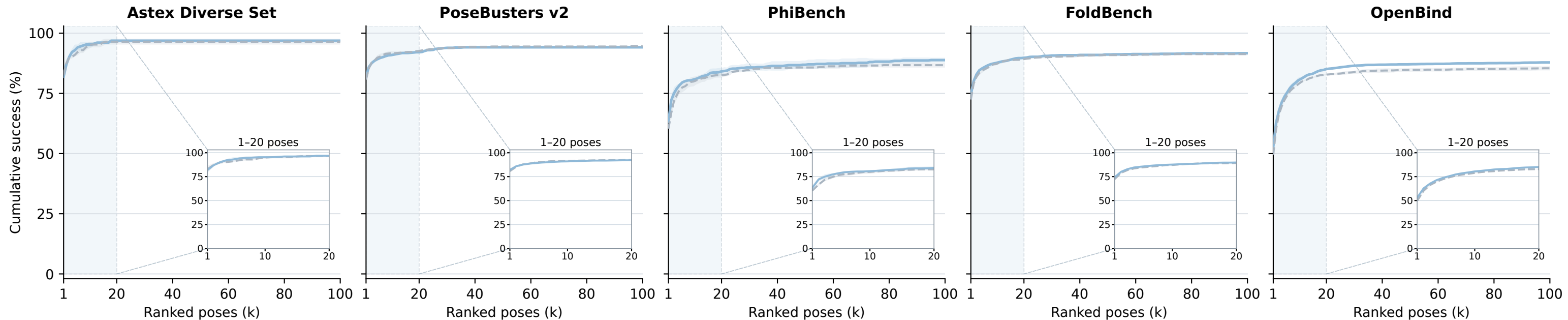


C Fragment count dependence

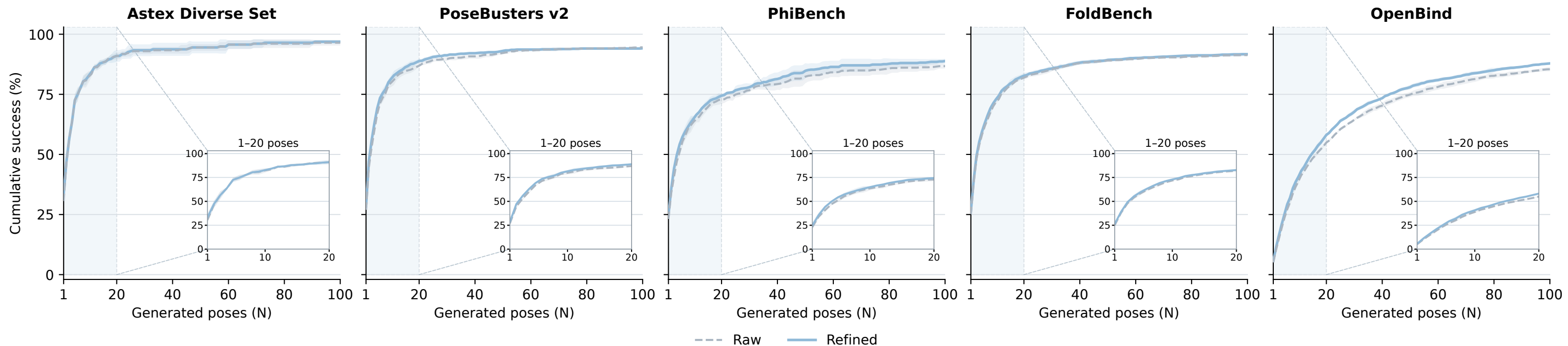


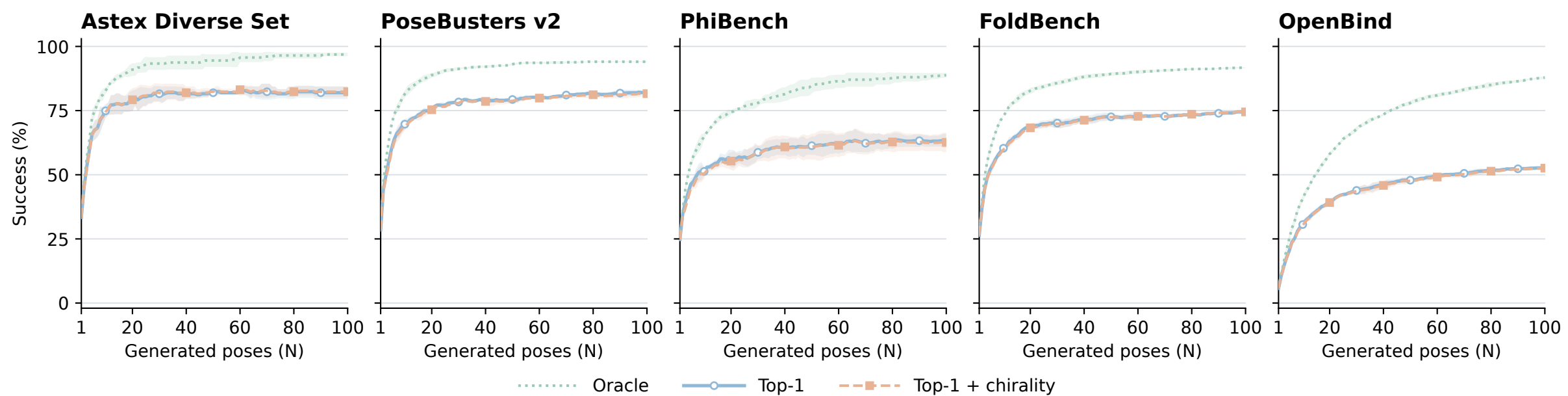
Oracle Top-1 Top-1 + PB-valid

A Confidence-ranked poses

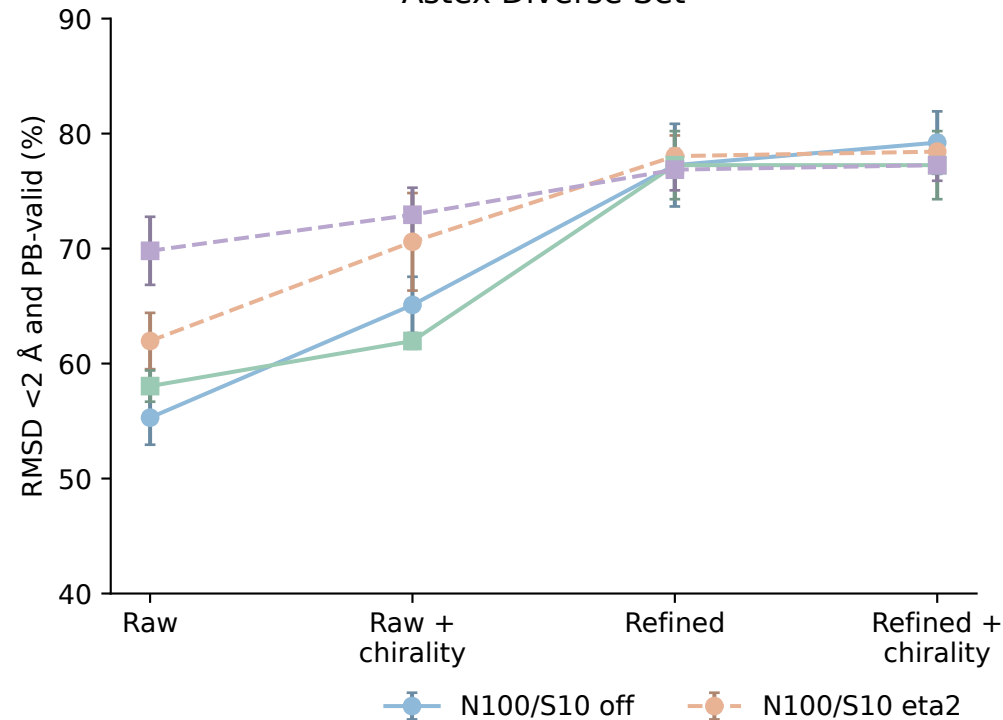


B Generation-order poses

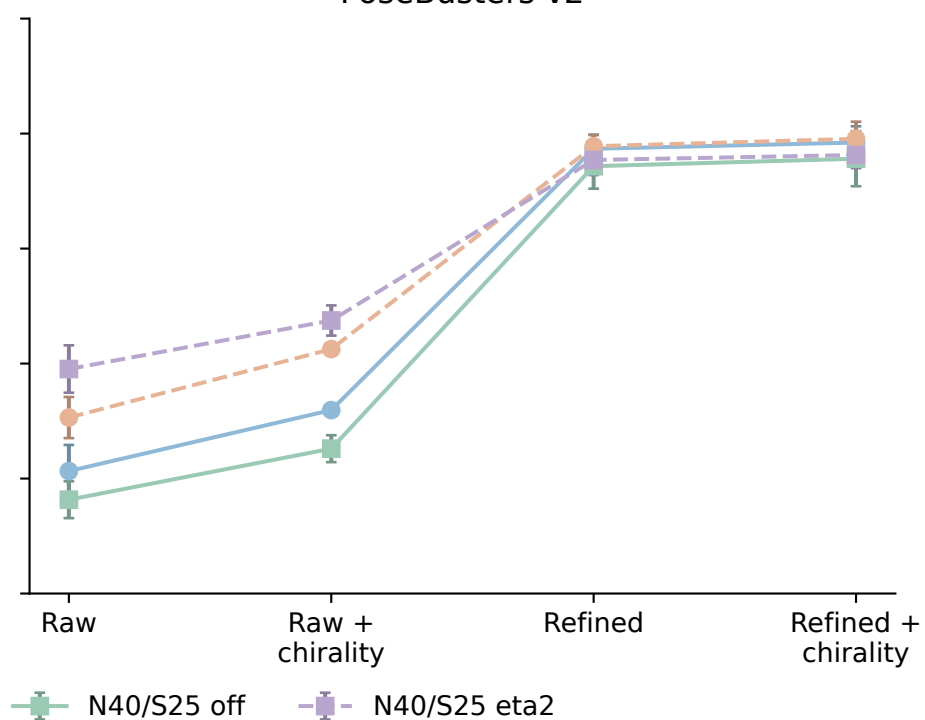


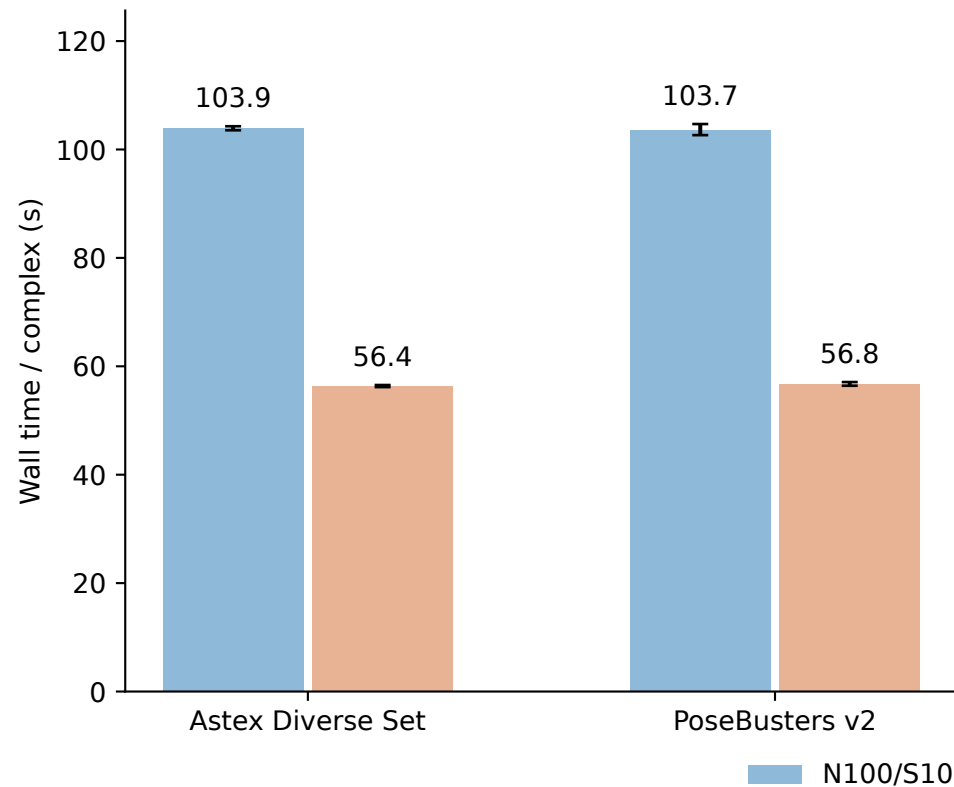
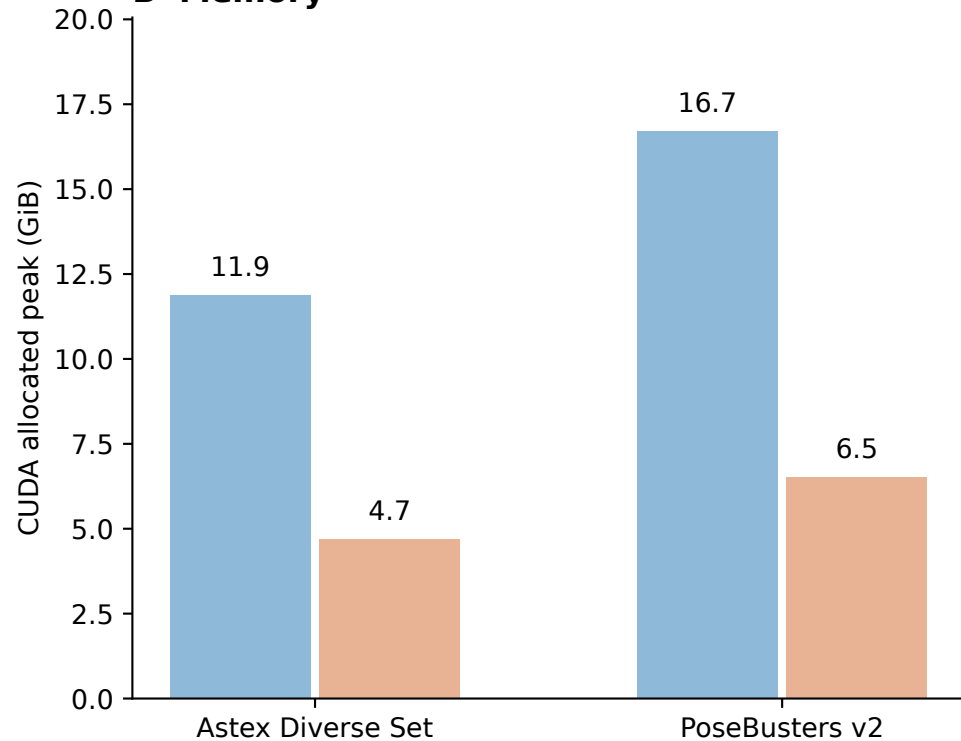


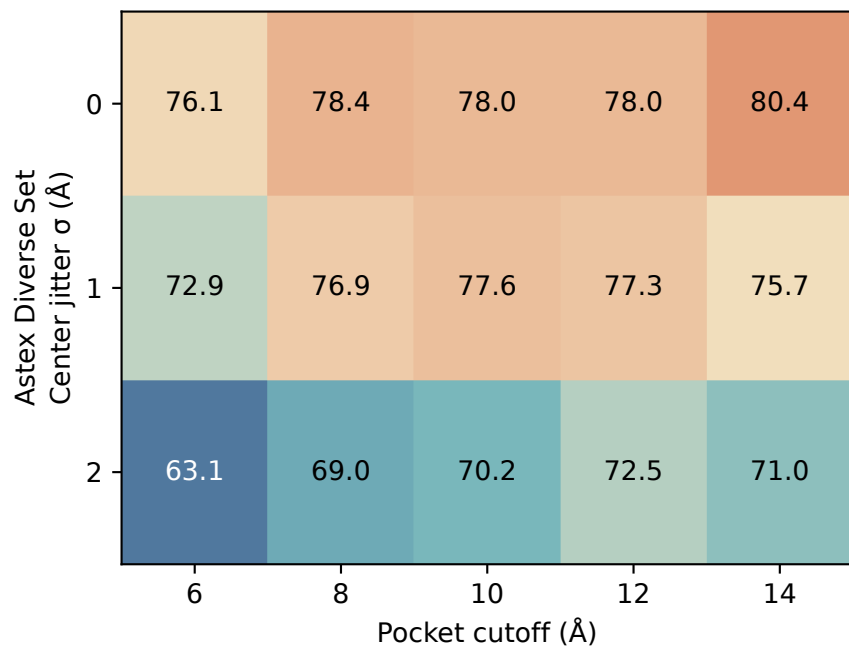
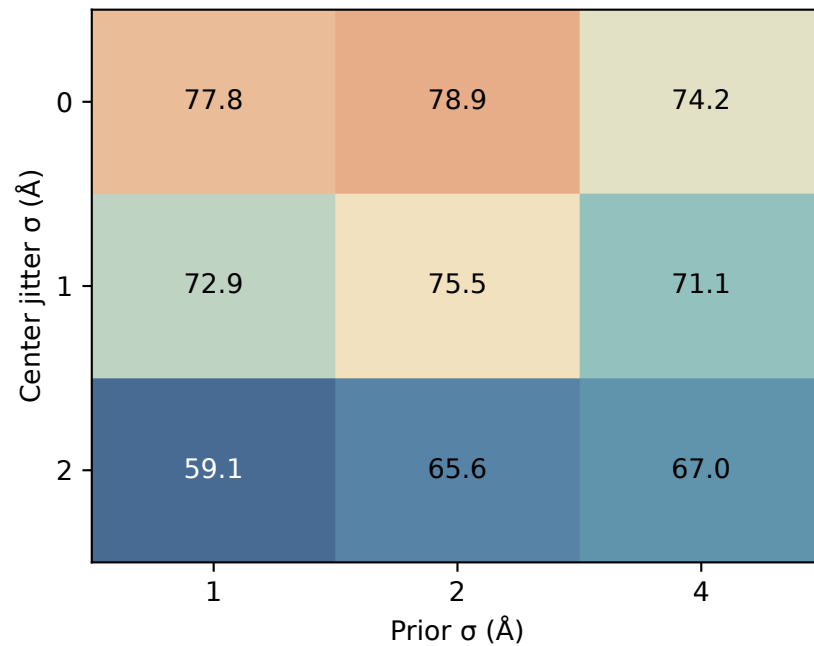
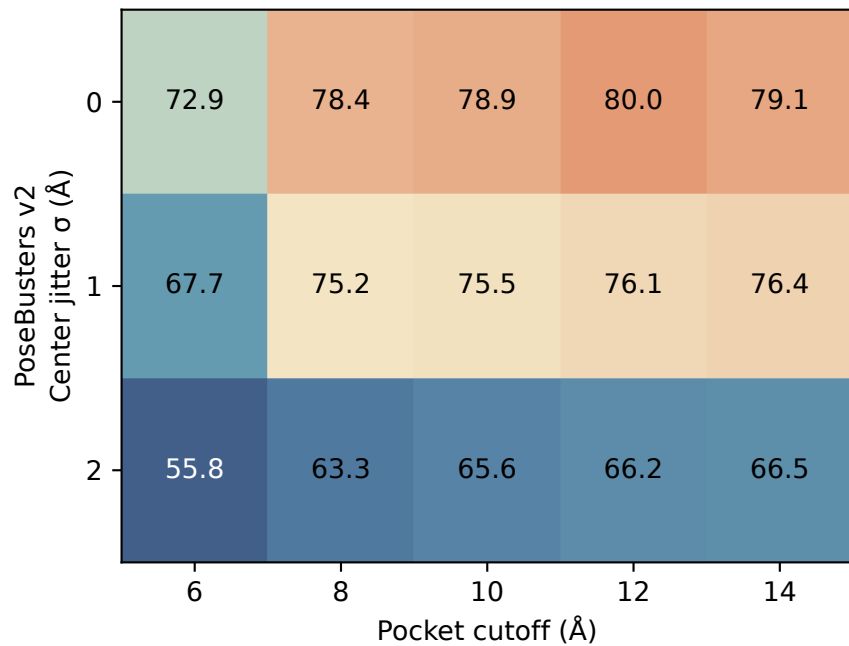
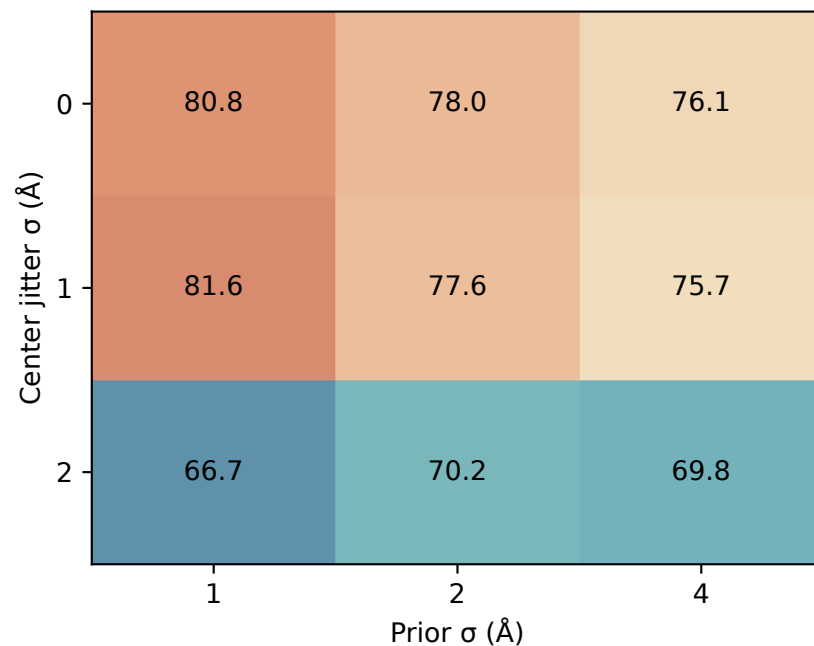
Astex Diverse Set



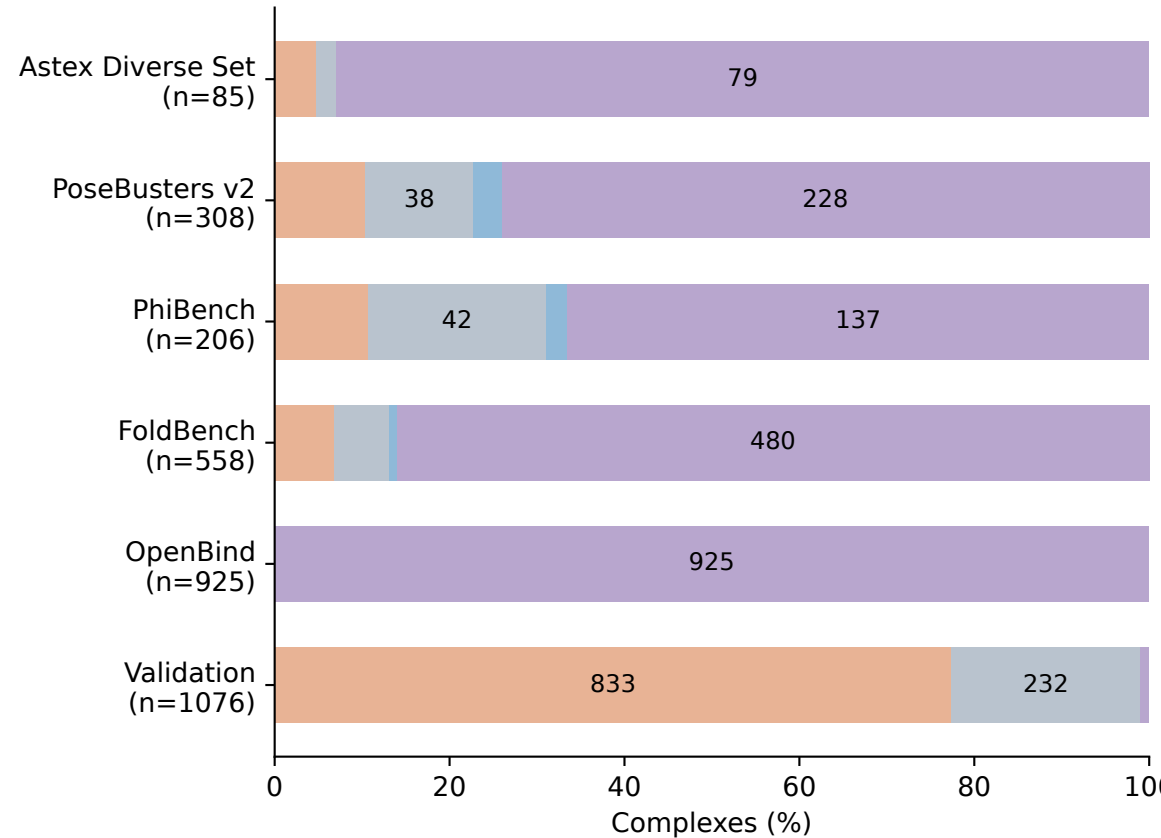
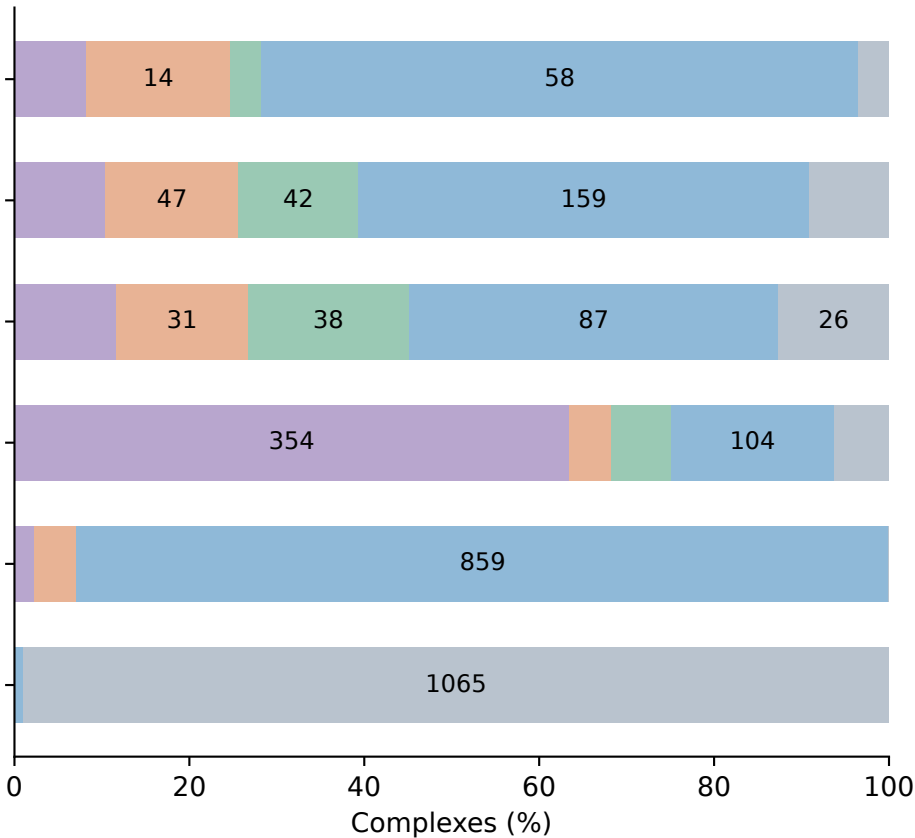
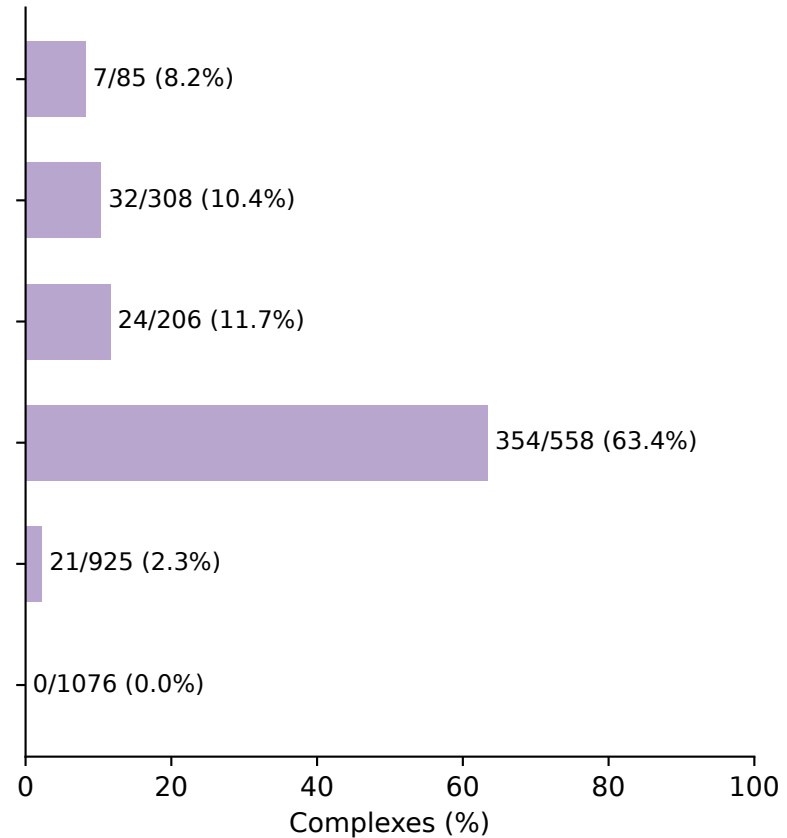
PoseBusters v2



A Runtime**B Memory**

A Pocket size**B Prior scale**

RMSD < 2 Å & PB-valid (%)

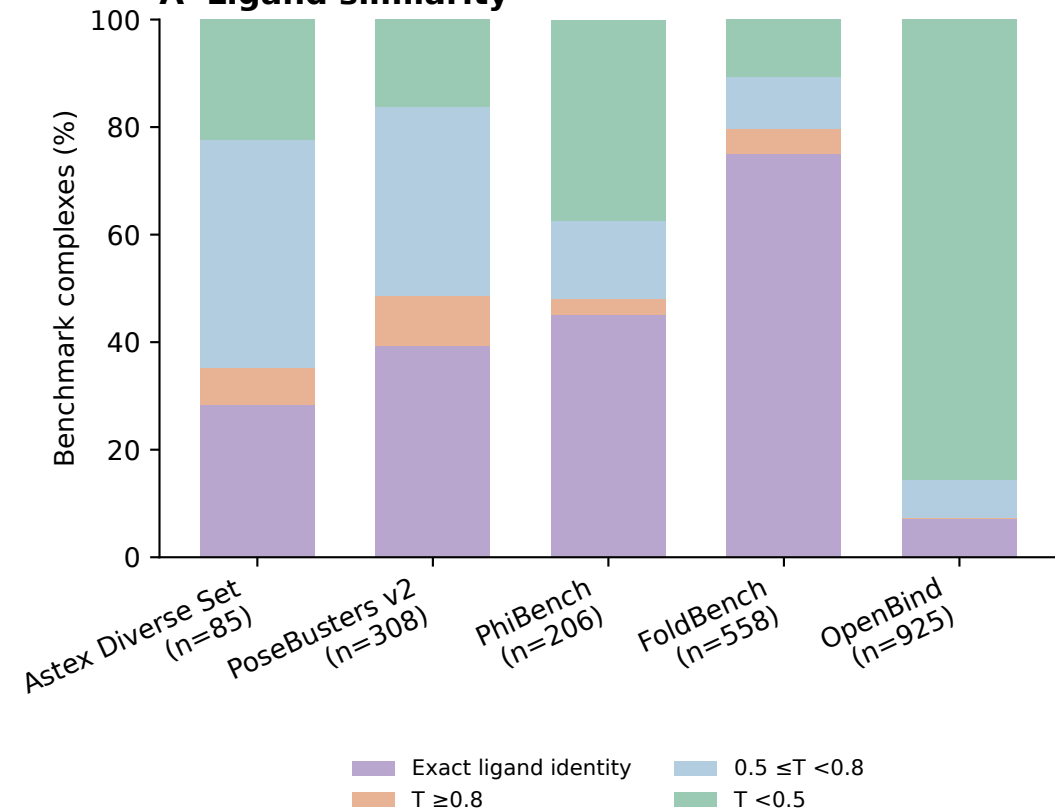
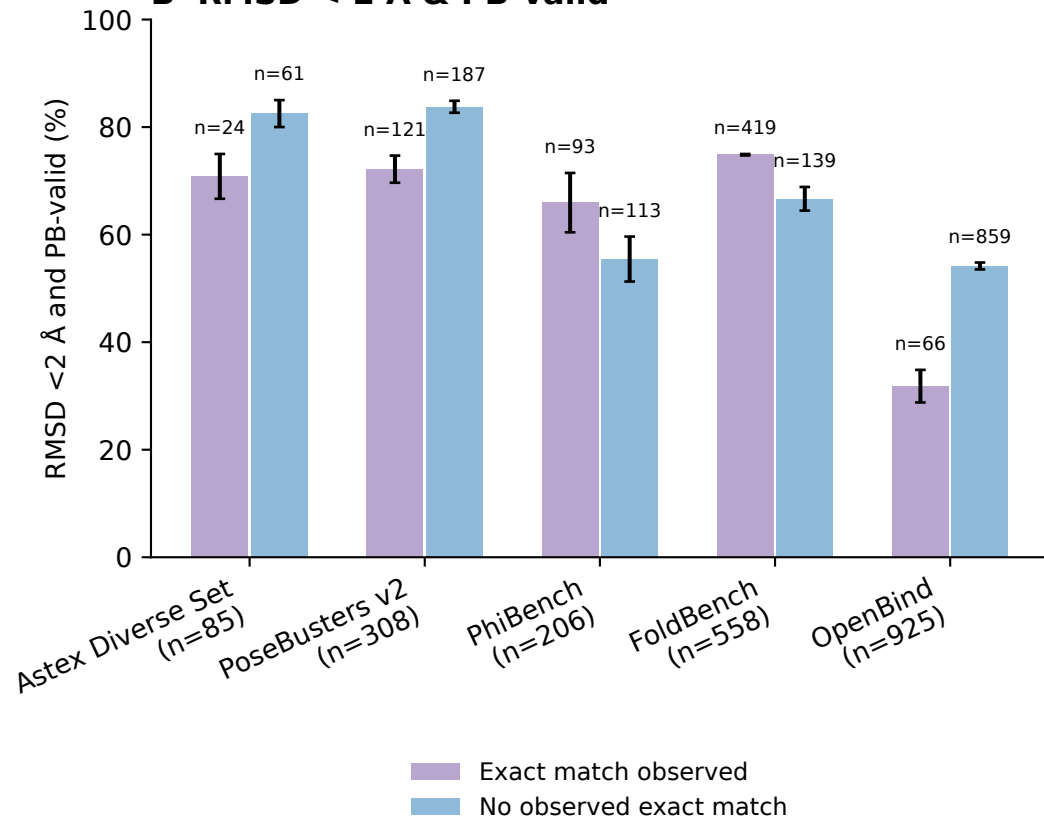
A Sequence identity**B Ligand-protein overlap****C Same training sample**

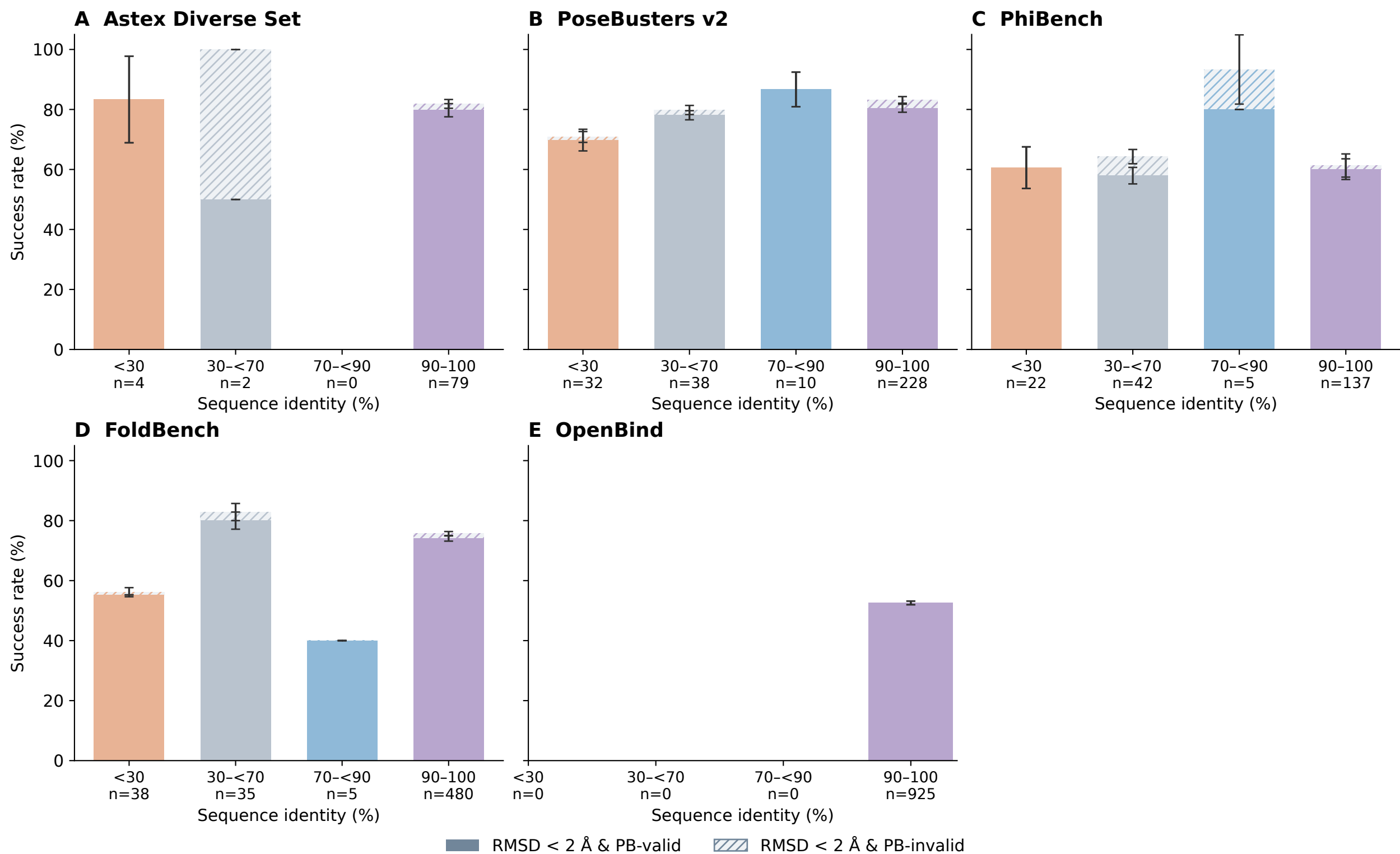
Sequence identity (%)

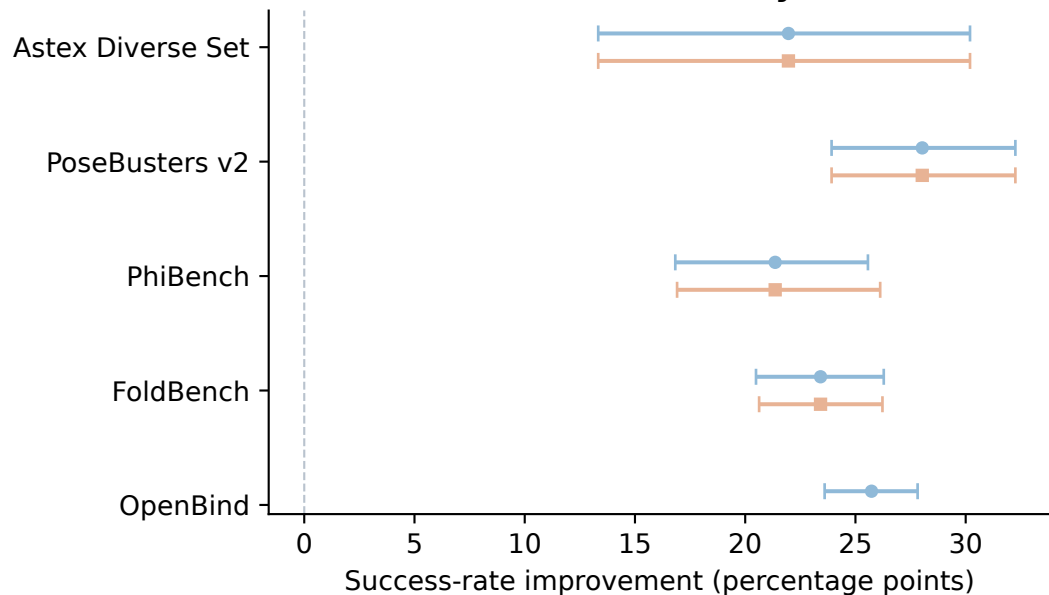
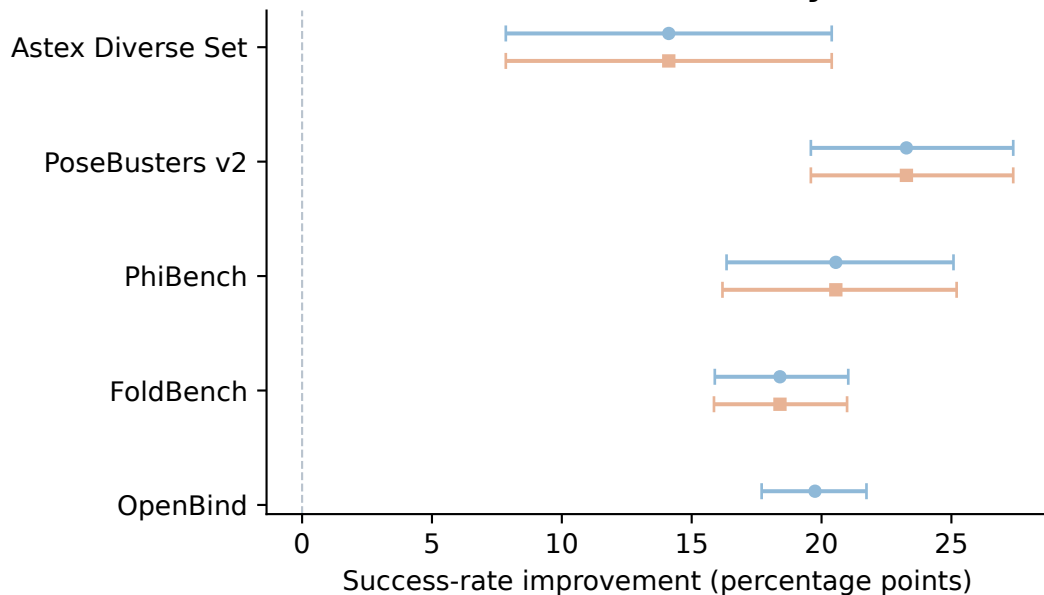
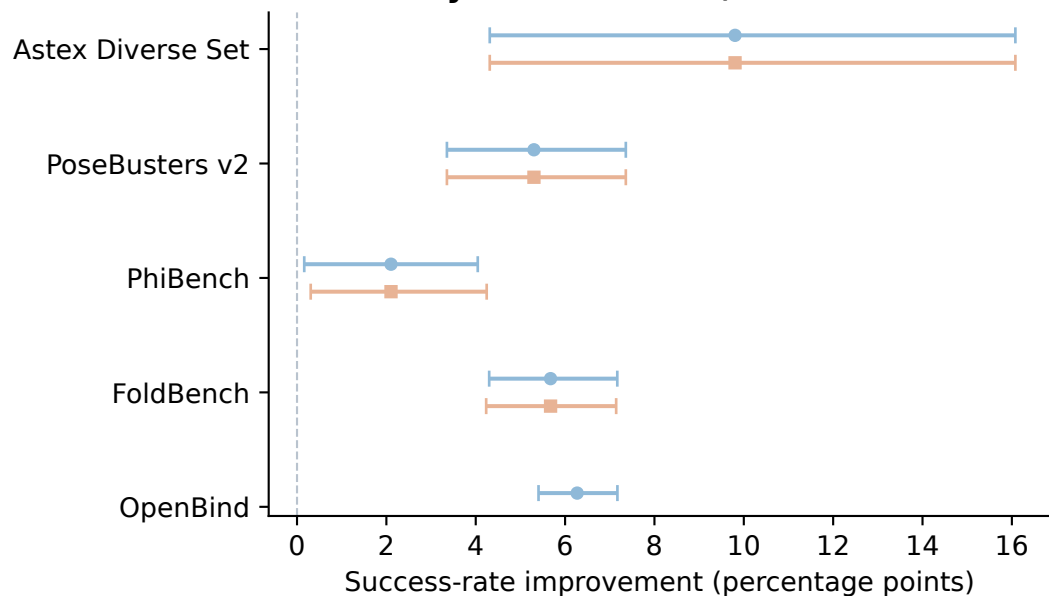
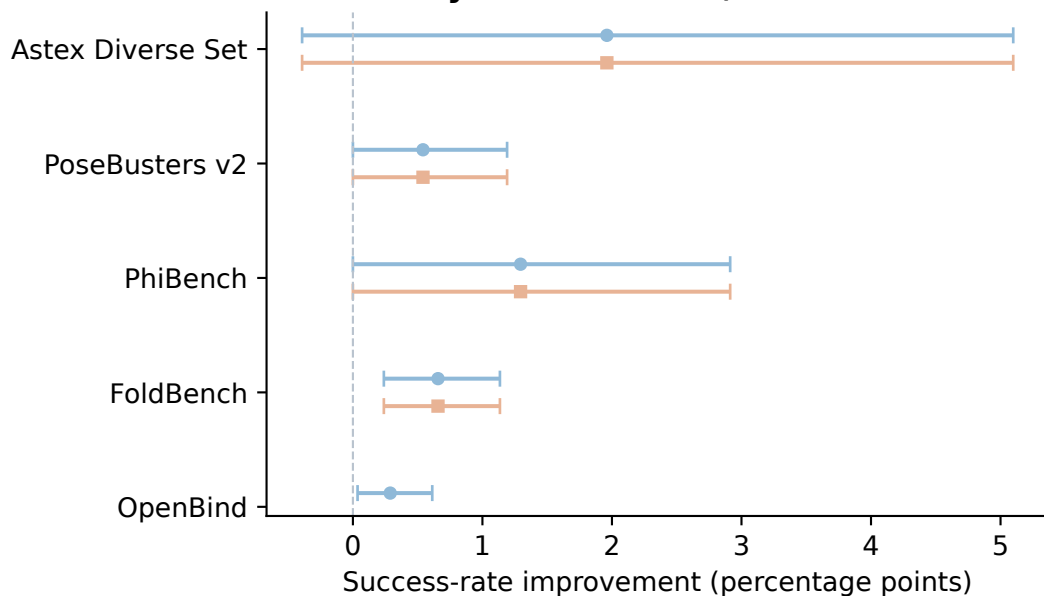
- <30%
- 30-<70%
- 70-<90%
- 90-100%

Ligand-protein overlap

- Same-sample overlap
- Separate matches
- Ligand only
- Sequence only
- Neither observed

A Ligand similarity**B RMSD < 2 Å & PB-valid**

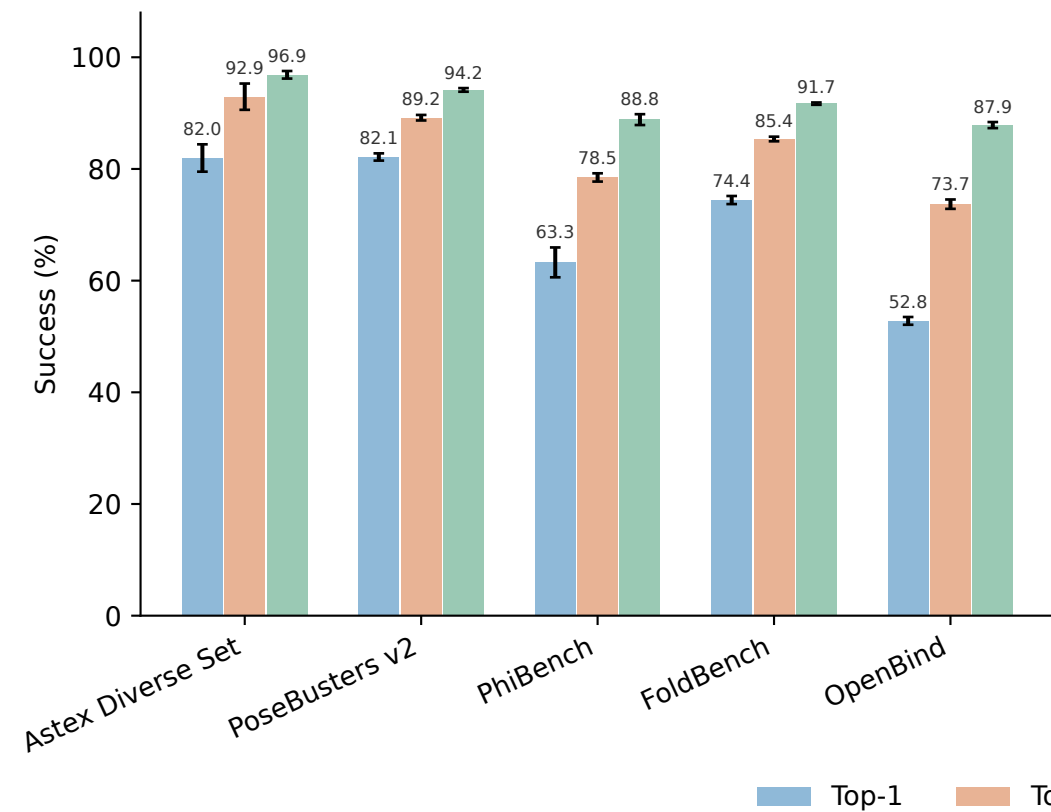
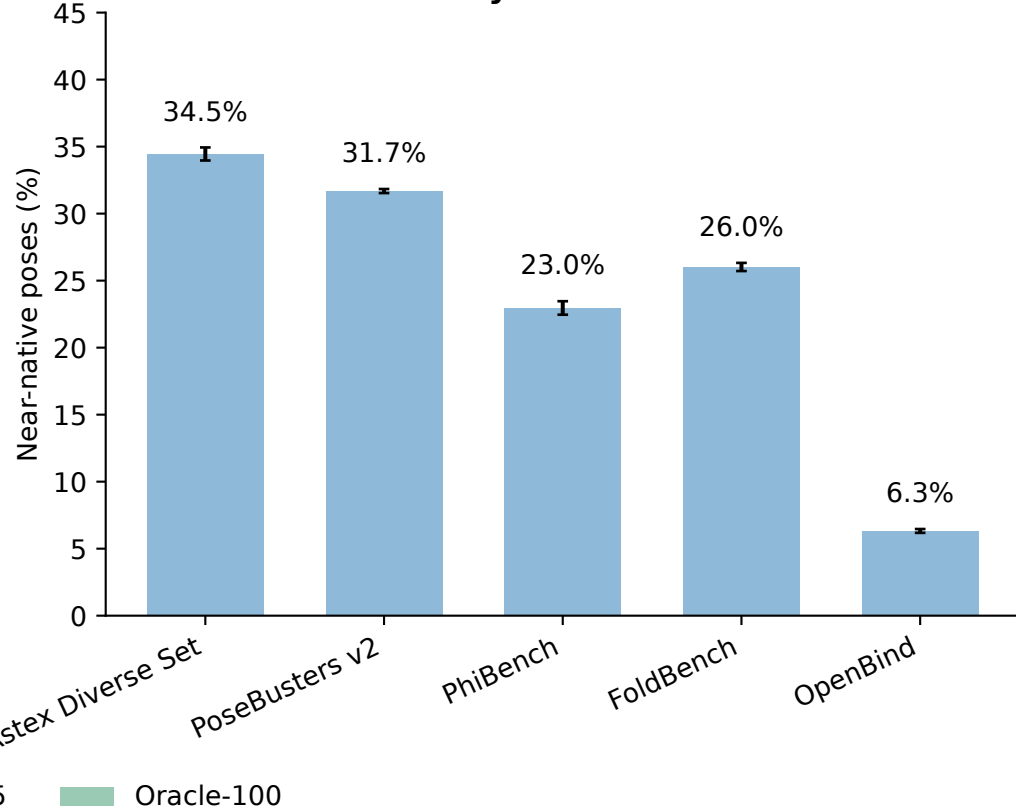


A Refined – raw, no chirality filter**B Refined – raw, with chirality filter****C Chirality filter off → on, raw****D Chirality filter off → on, refined**

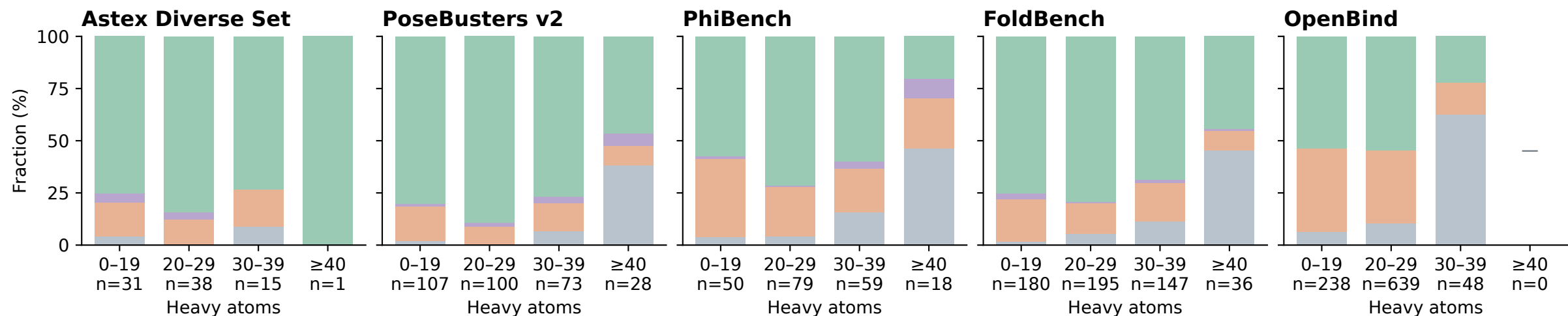
Complex bootstrap



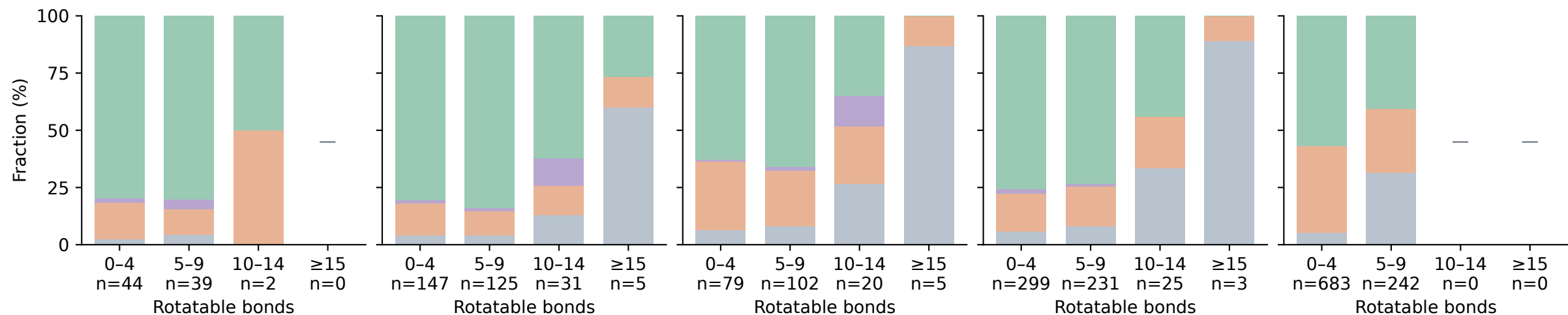
PDB-cluster bootstrap

A Selection**B Near-native density**

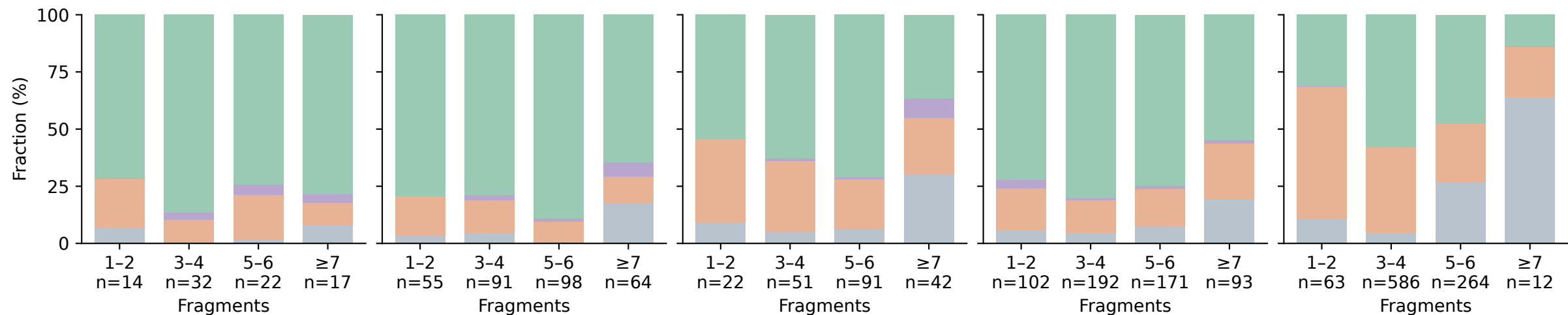
A Ligand size dependence



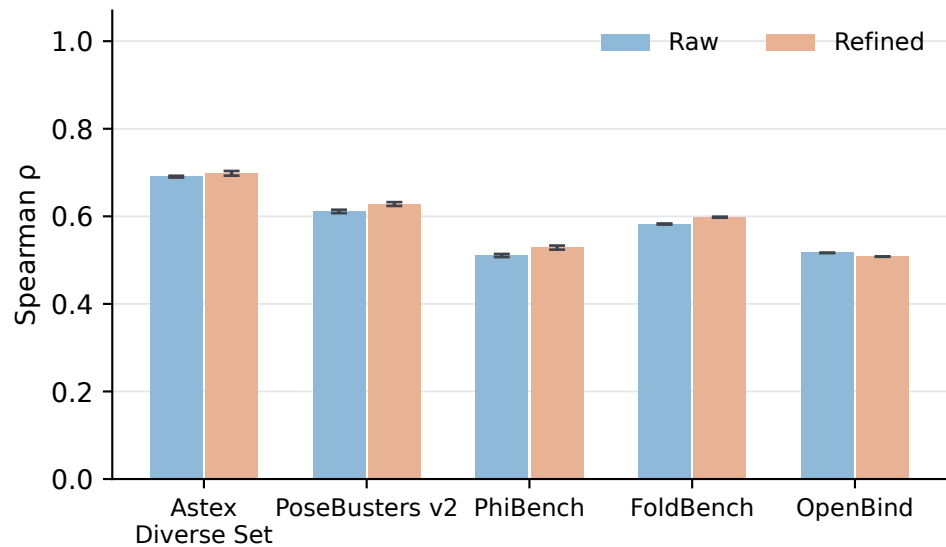
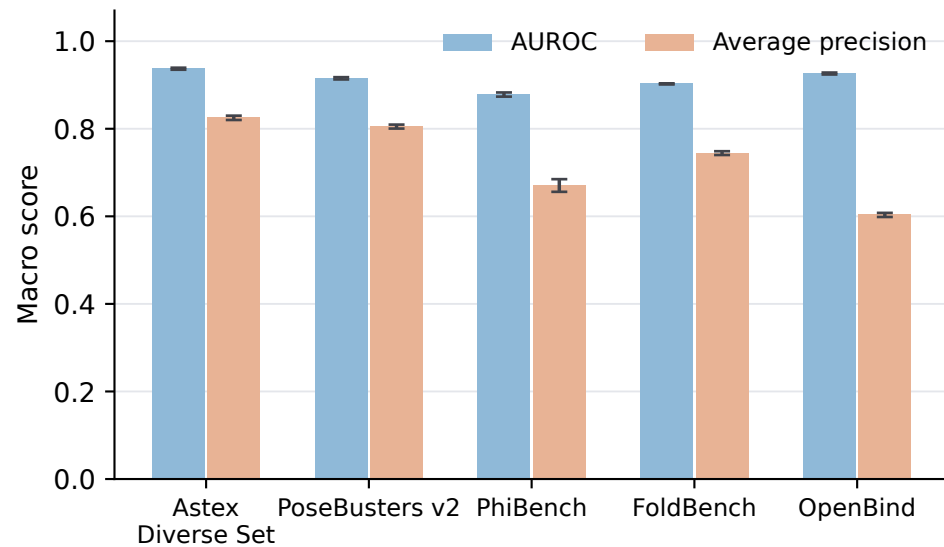
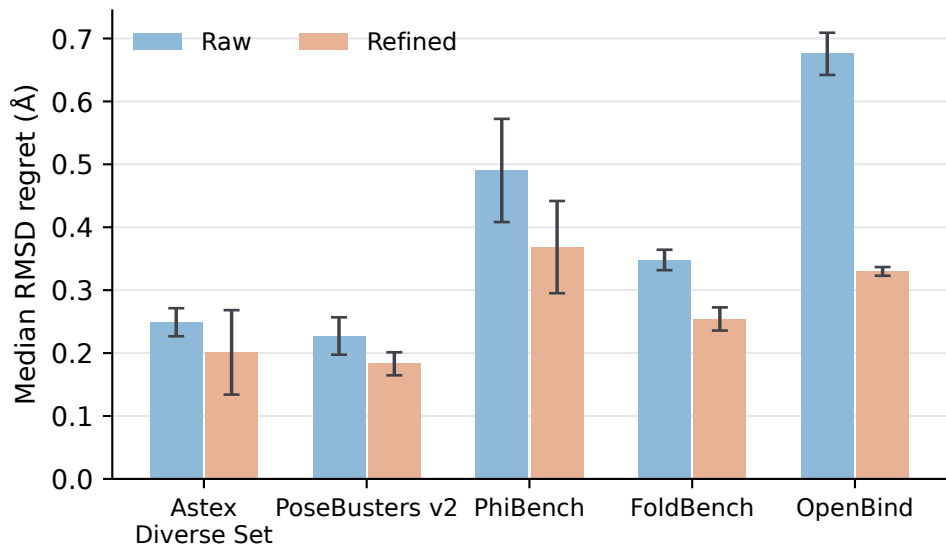
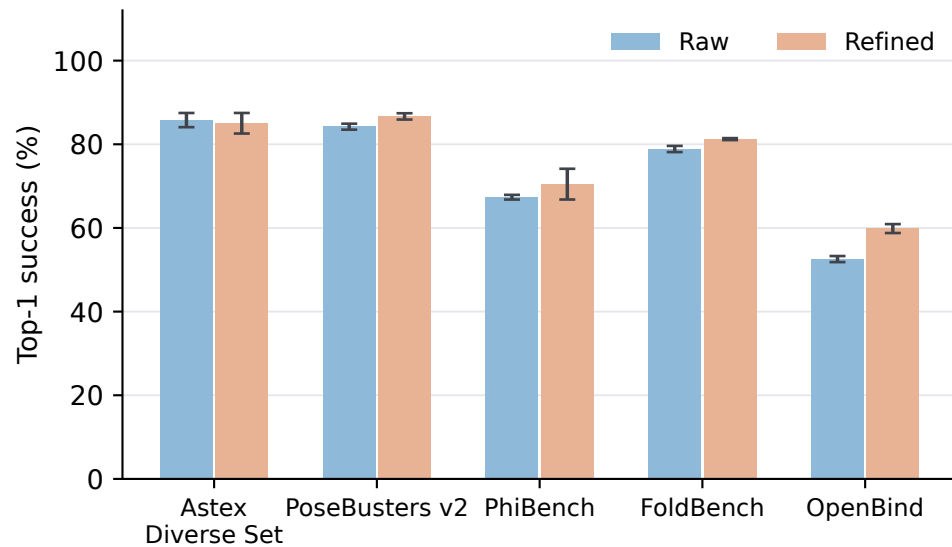
B Rotatable bond dependence

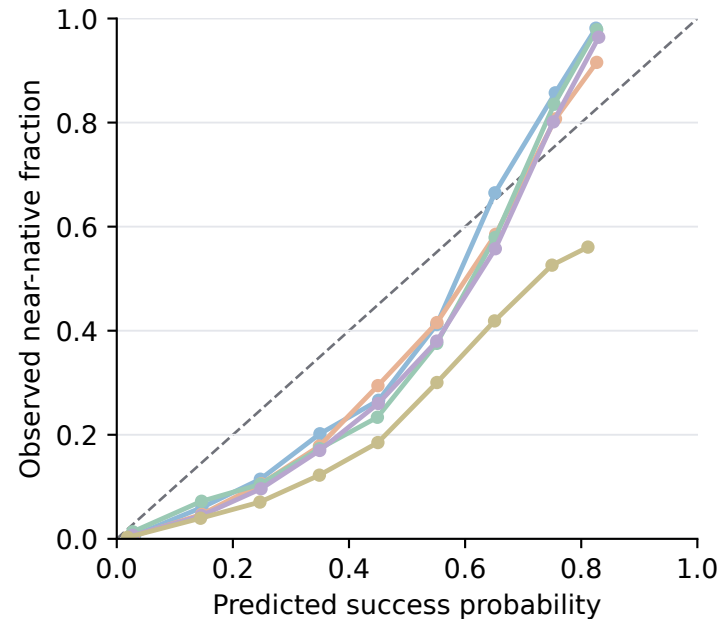
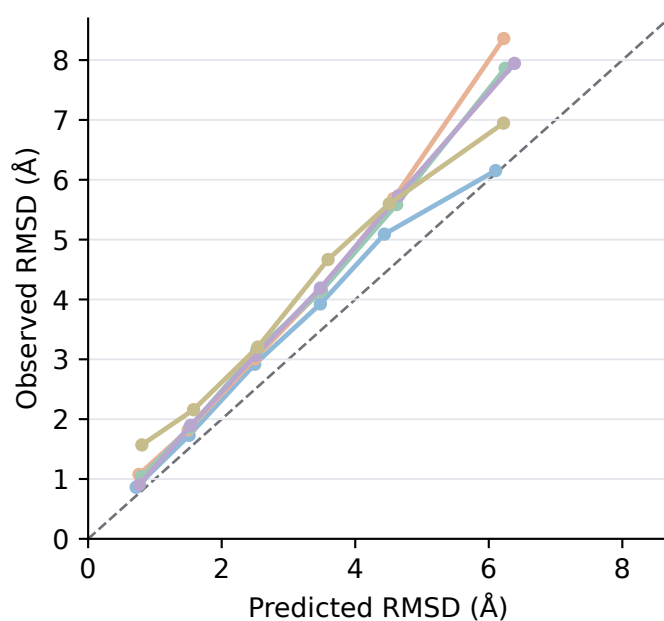
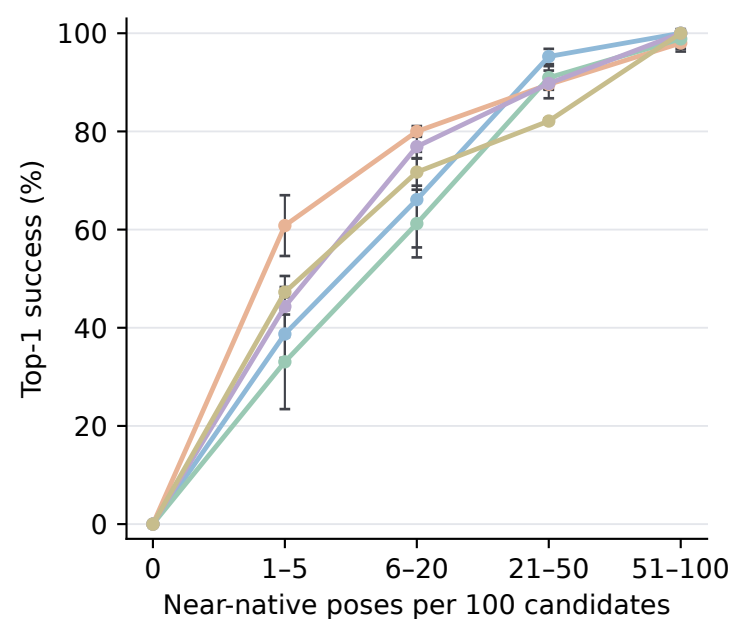


C Fragment count dependence

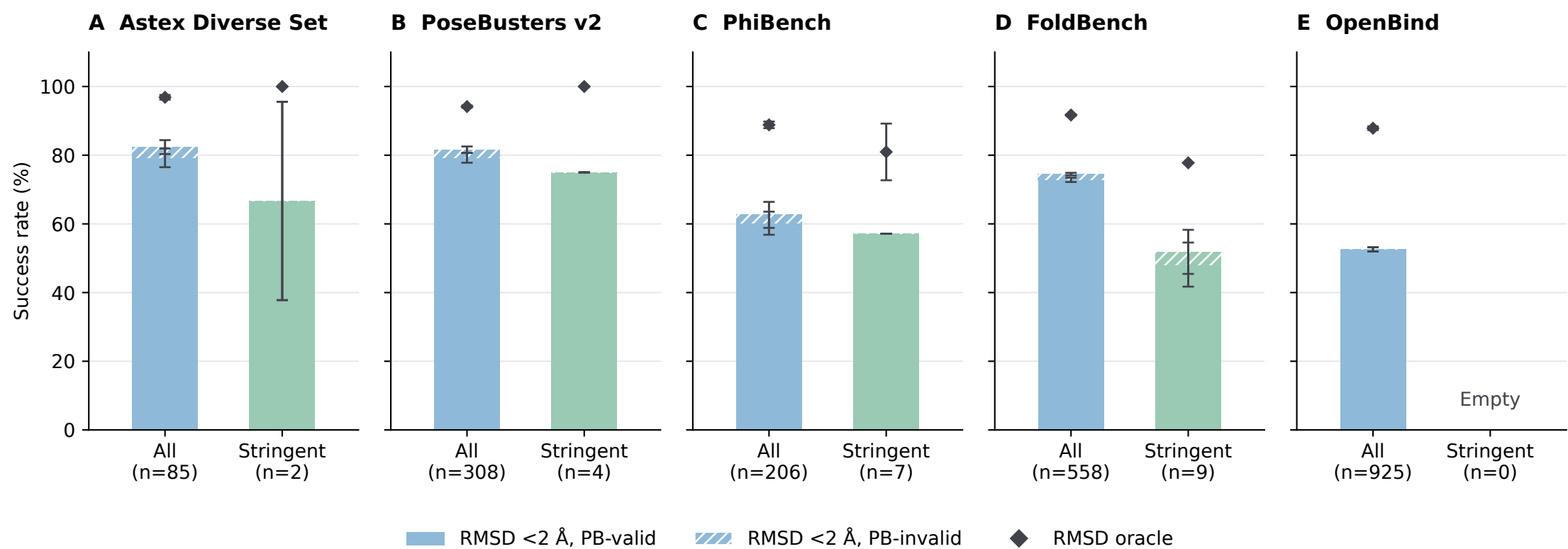


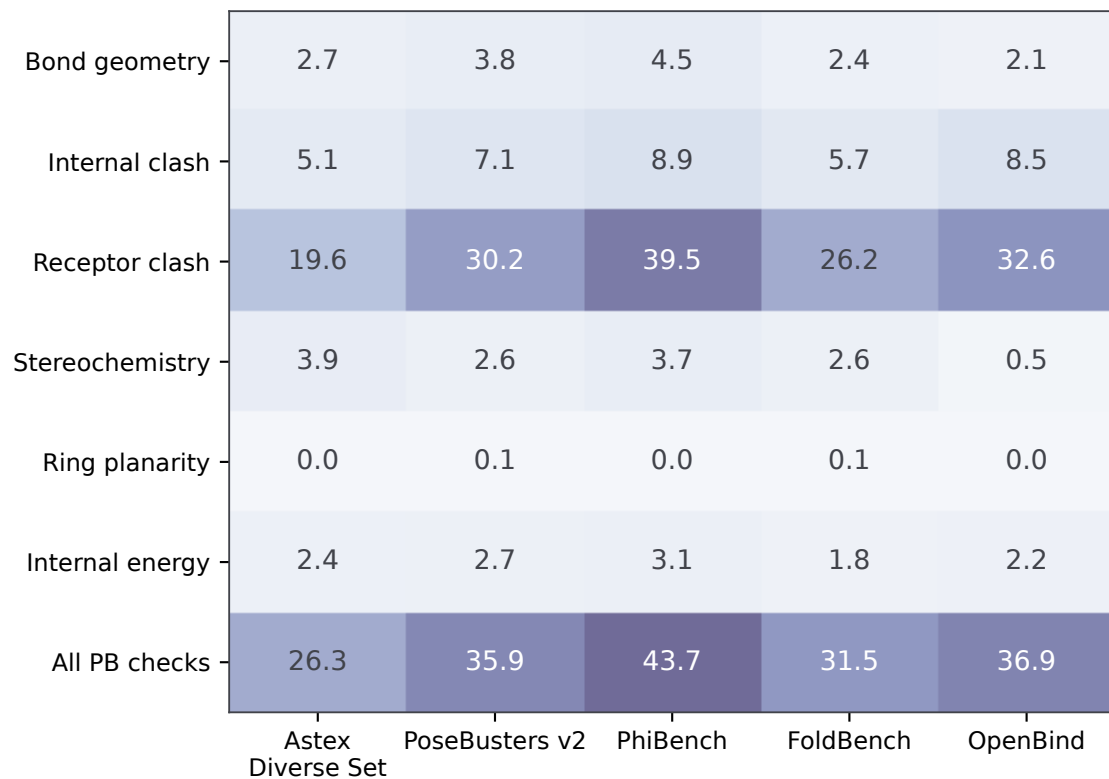
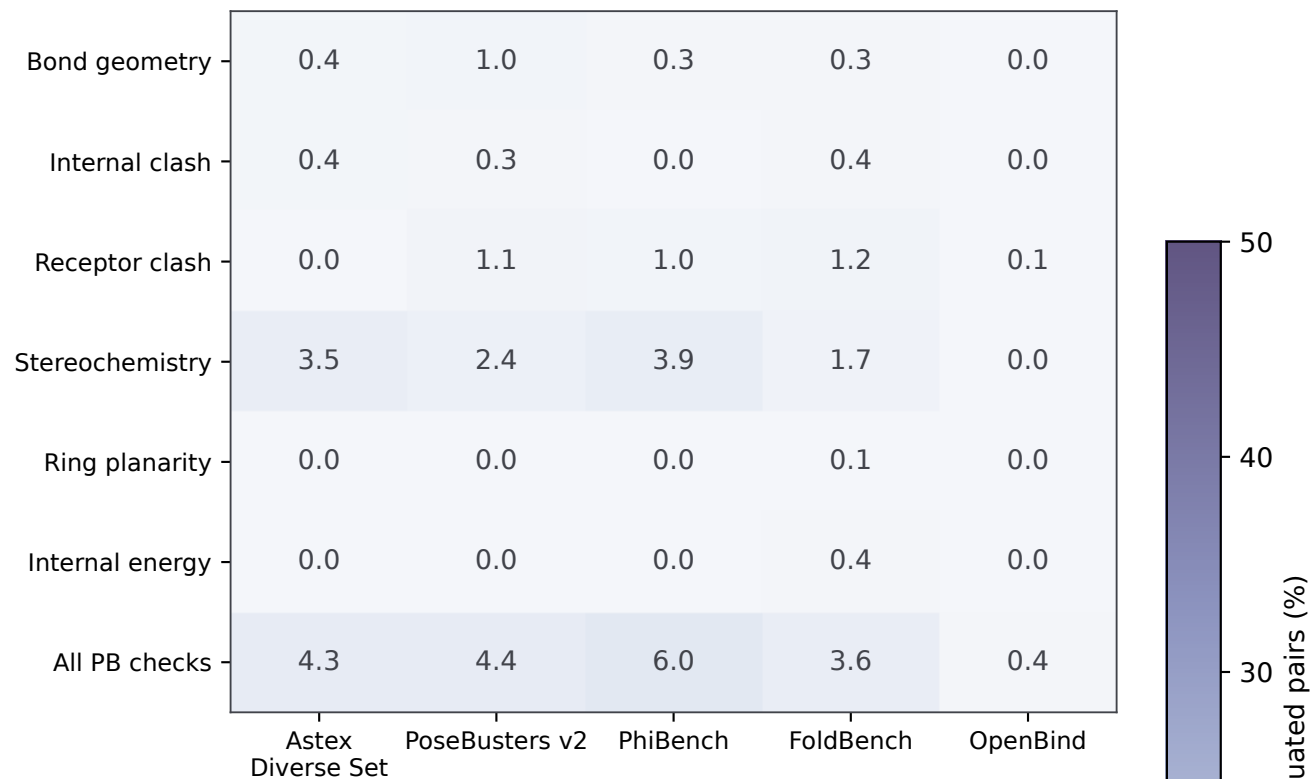
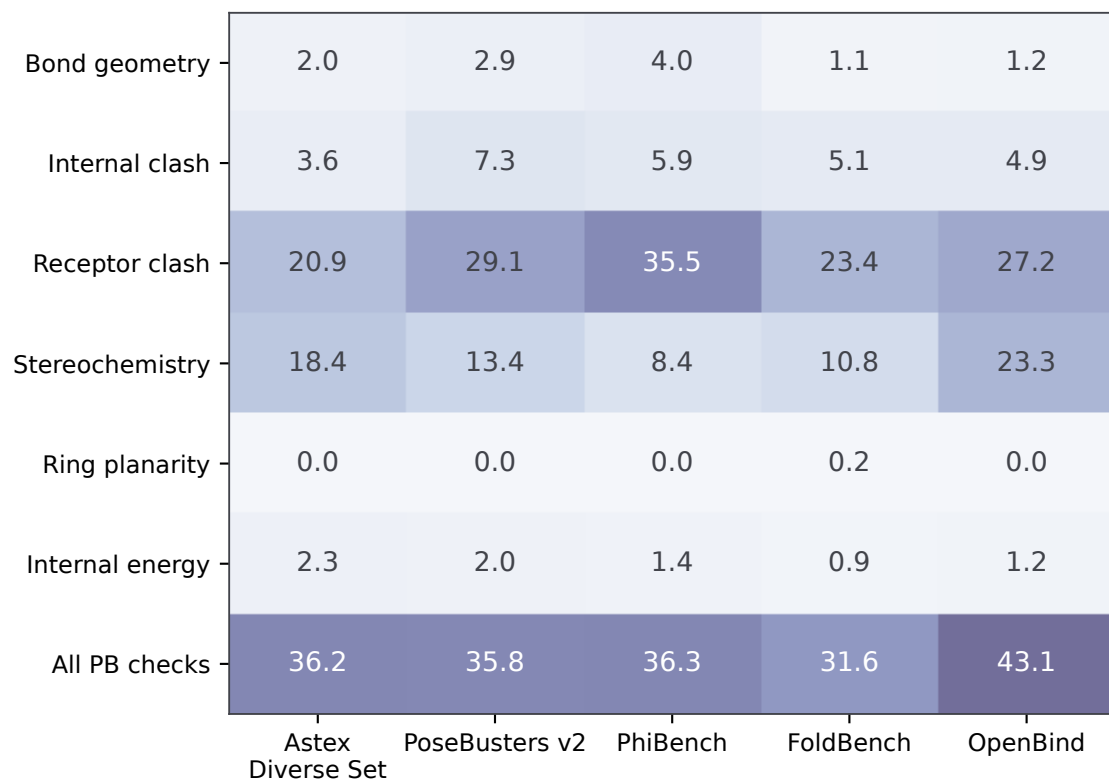
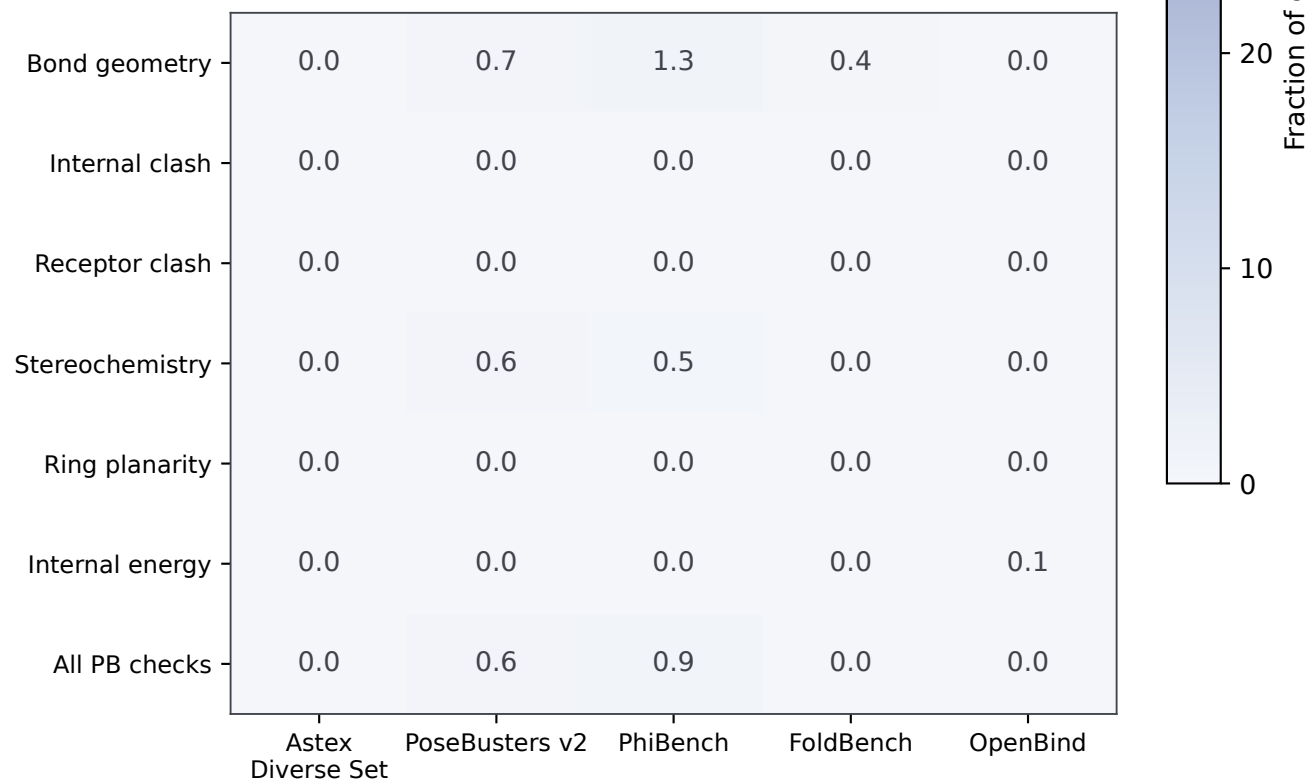
■ Near-native pose generation failure ■ Selection failure ■ PB failure ■ Success

A Within-bank ranking**B Near-native discrimination****C Selection regret****D Selection given a successful candidate**

A Success-probability reliability**B RMSD prediction reliability****C Near-native candidate density**

—●— Astex Diverse Set —●— PoseBusters v2 —●— PhiBench —●— FoldBench —●— OpenBind



A Raw selected-pose failure**B Refined selected-pose failure****C Matched candidate: fail → pass****D Matched candidate: pass → fail**

Fraction of evaluated pairs (%)

50

40

30

20

10

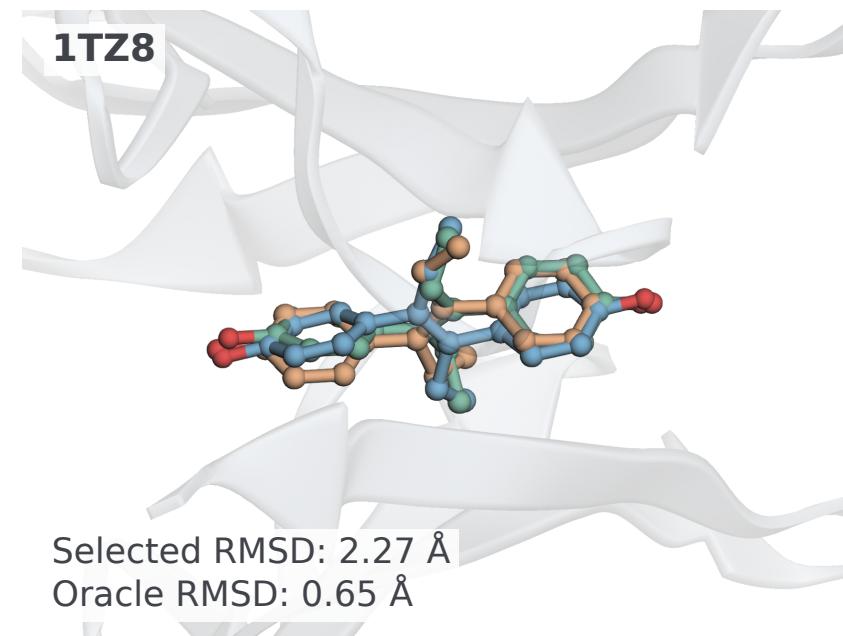
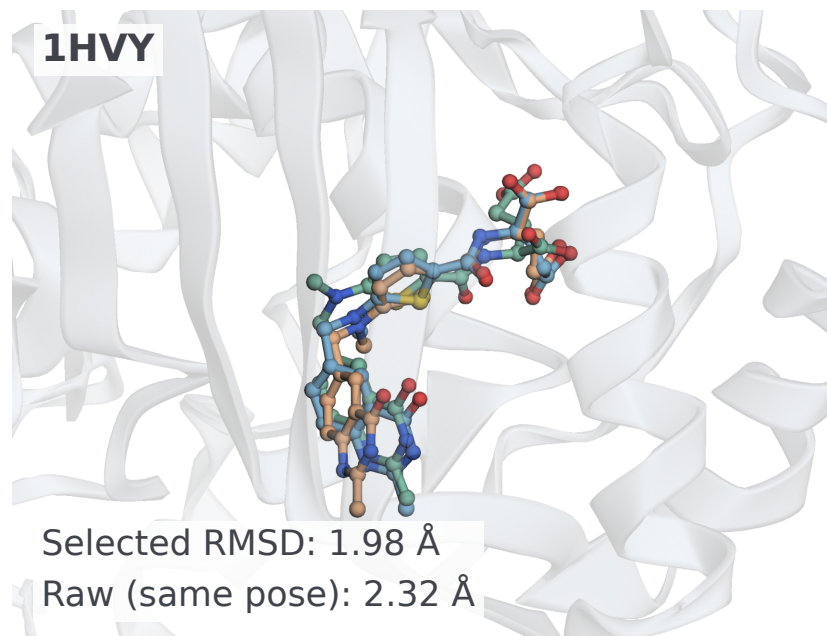
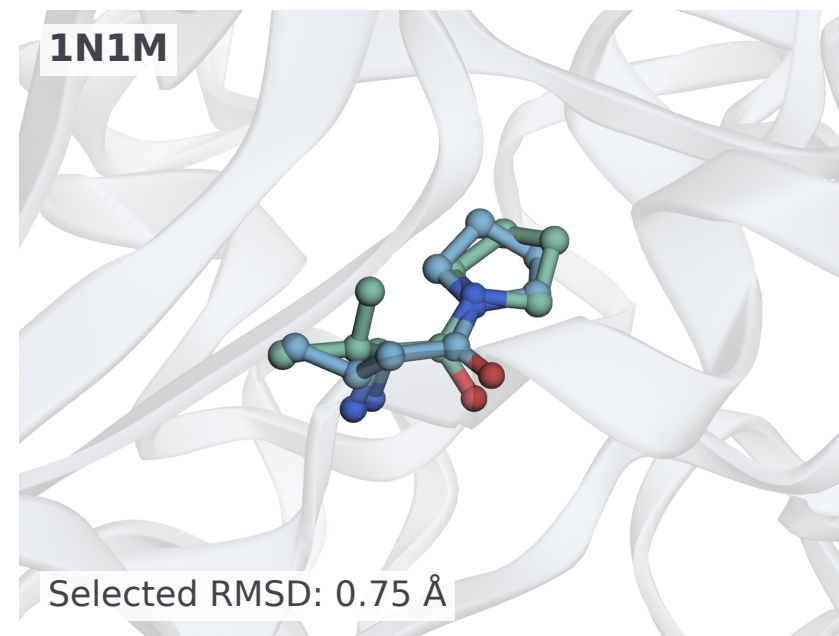
0

Successful pose

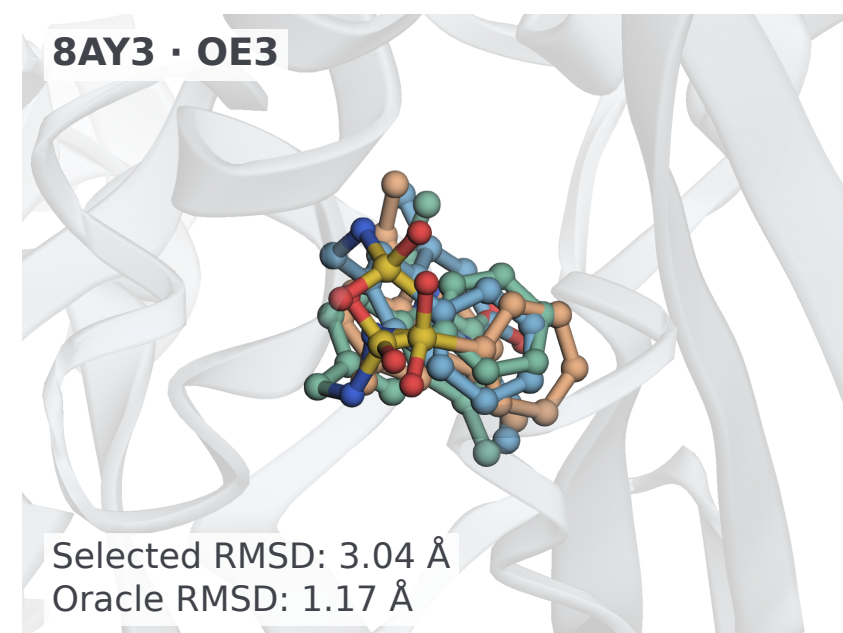
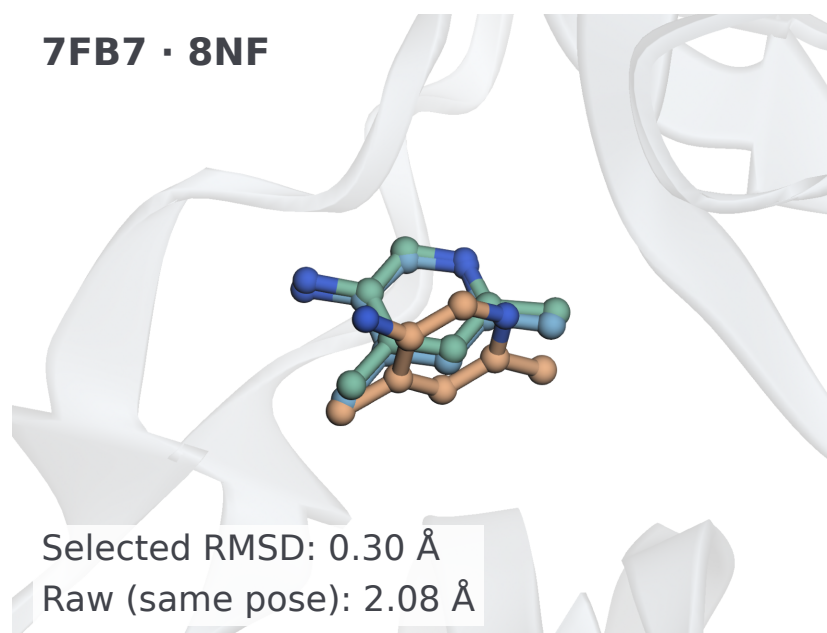
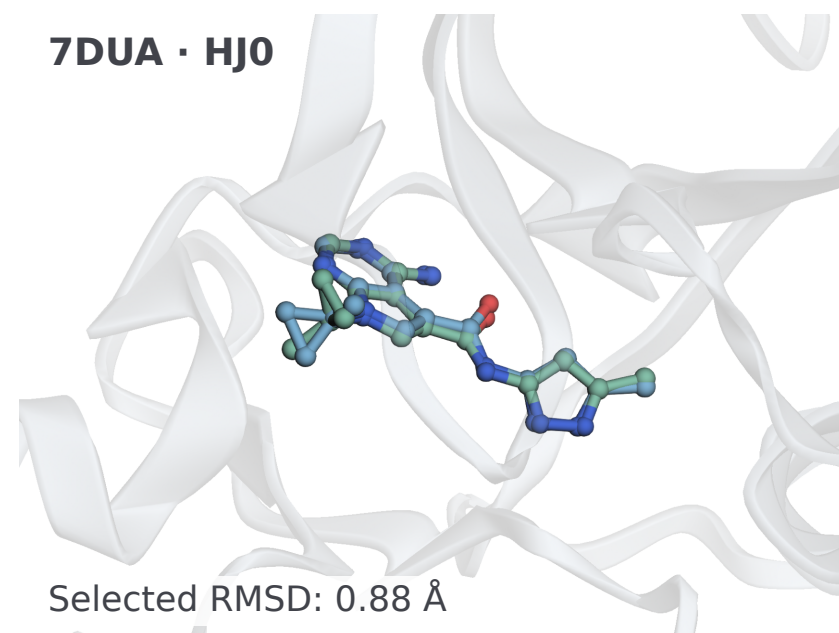
Refinement rescue

Selection failure

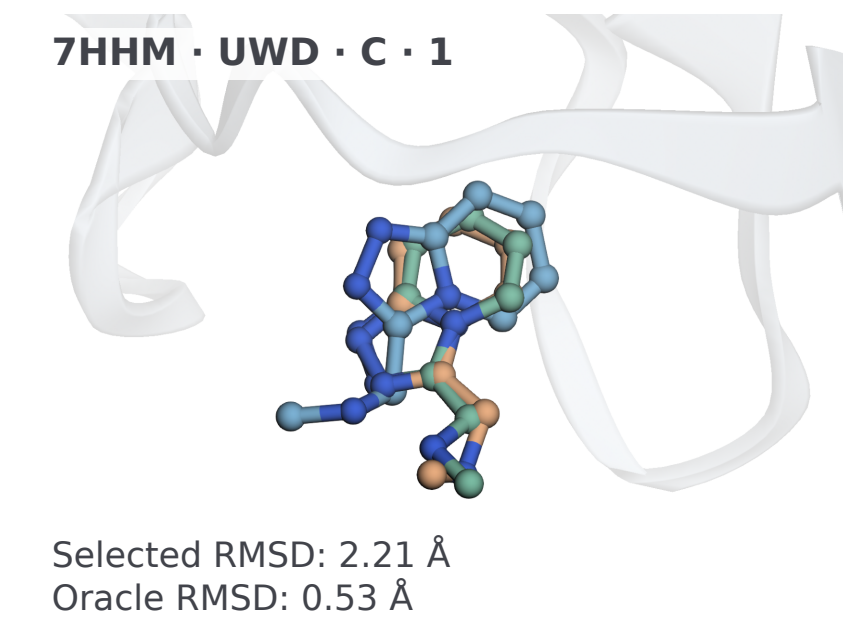
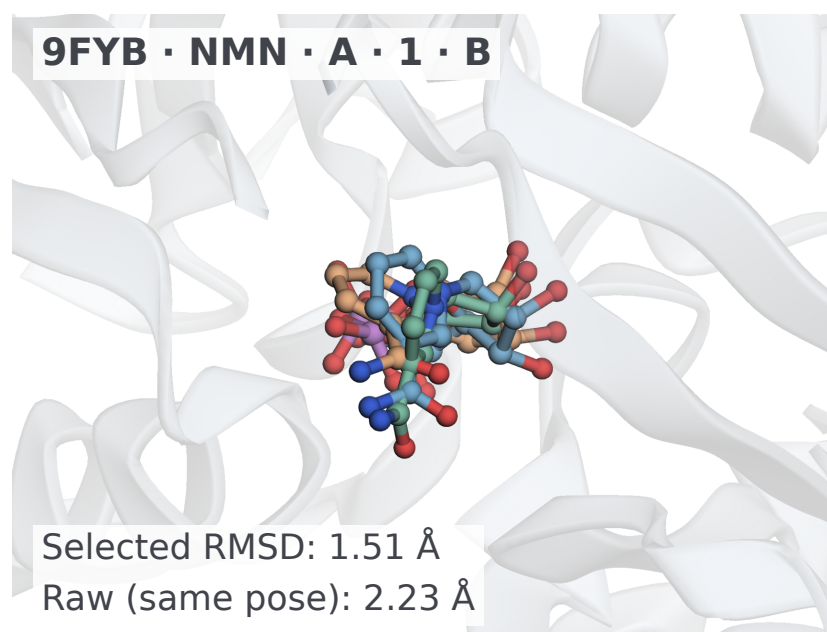
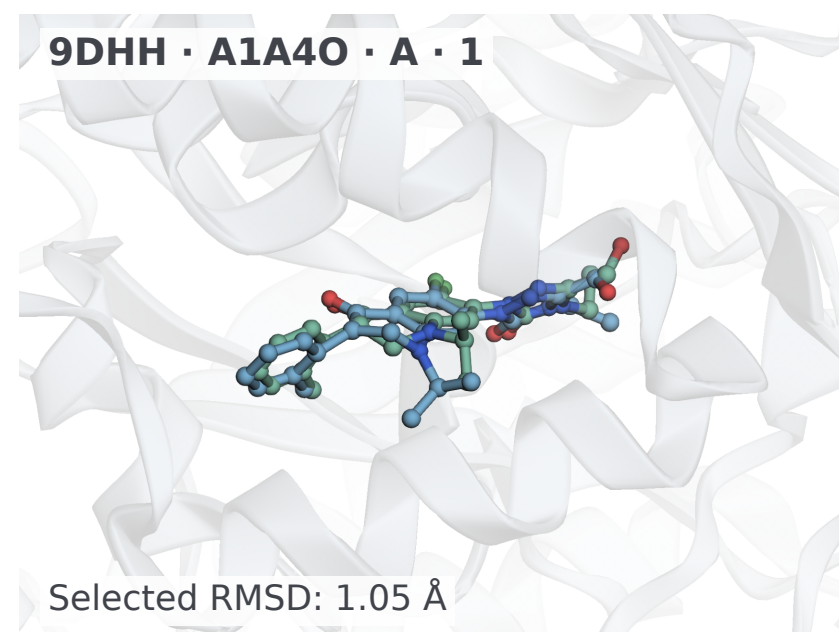
A Astex Diverse Set



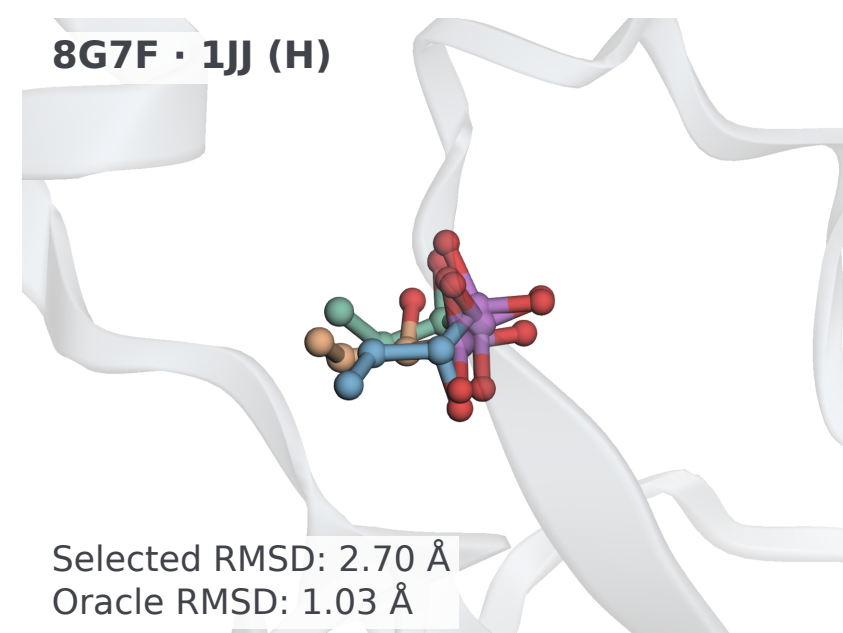
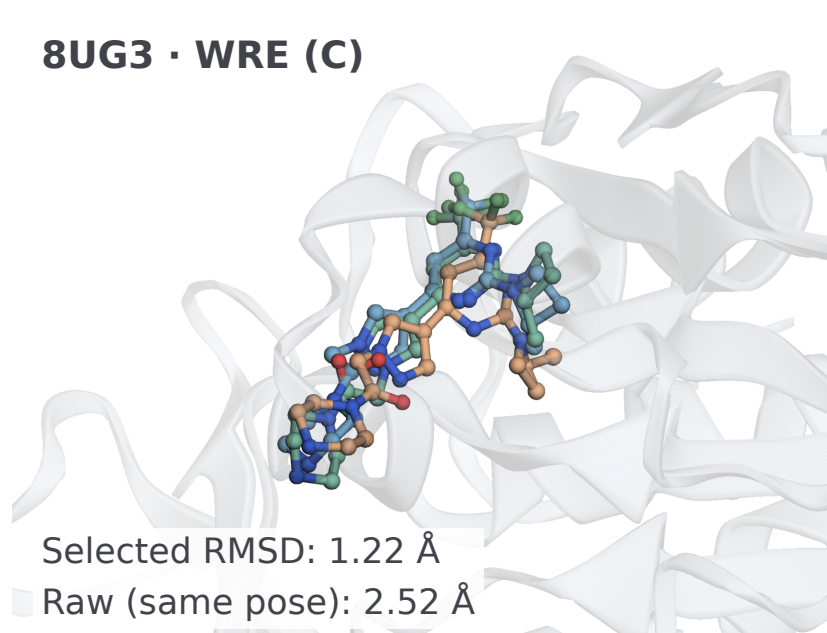
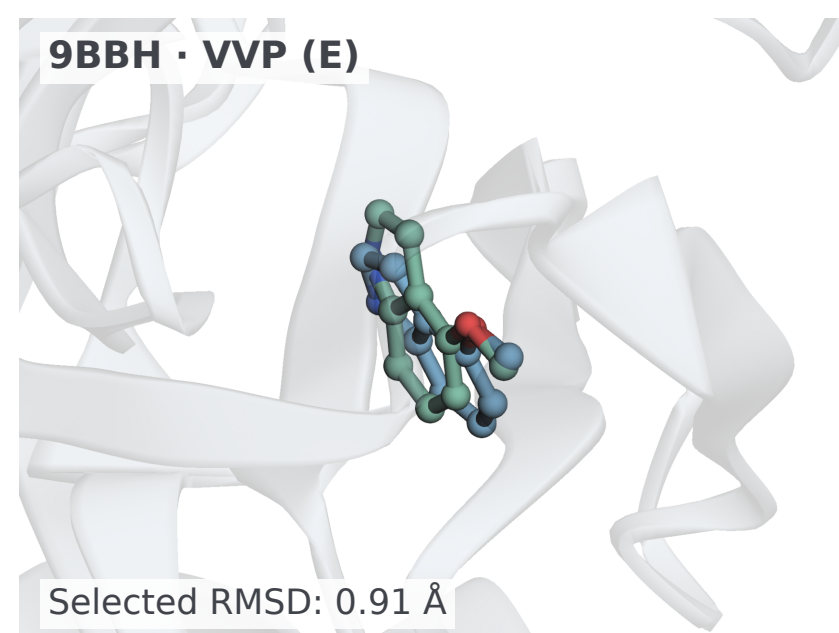
B PoseBusters v2



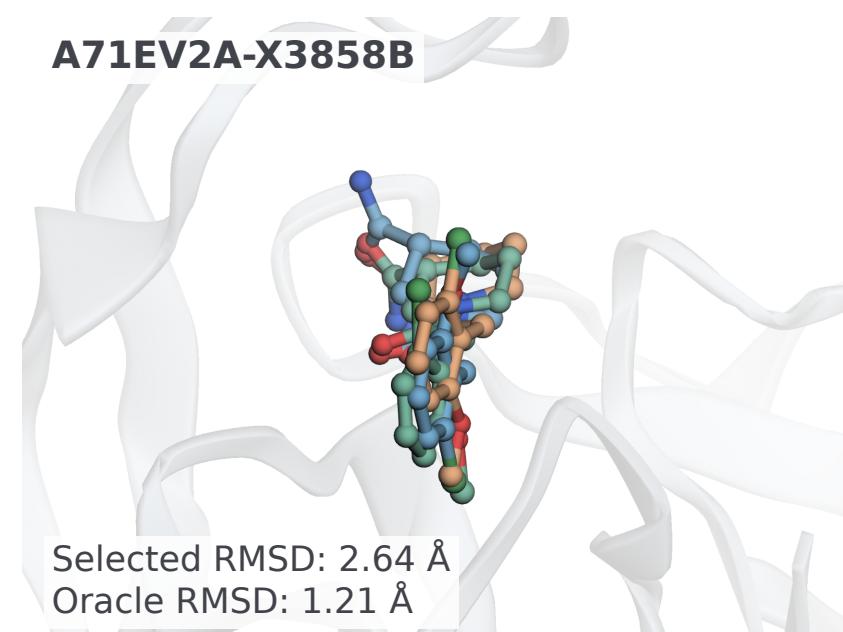
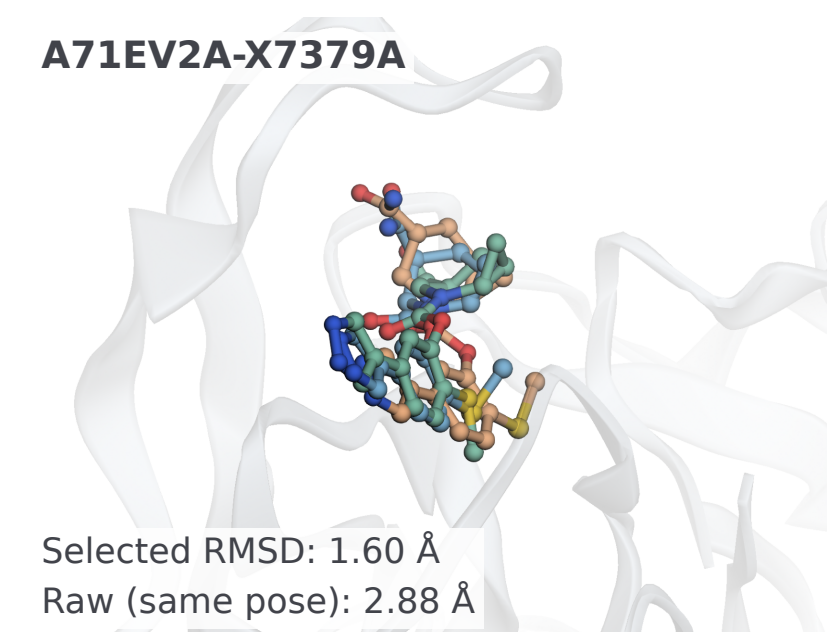
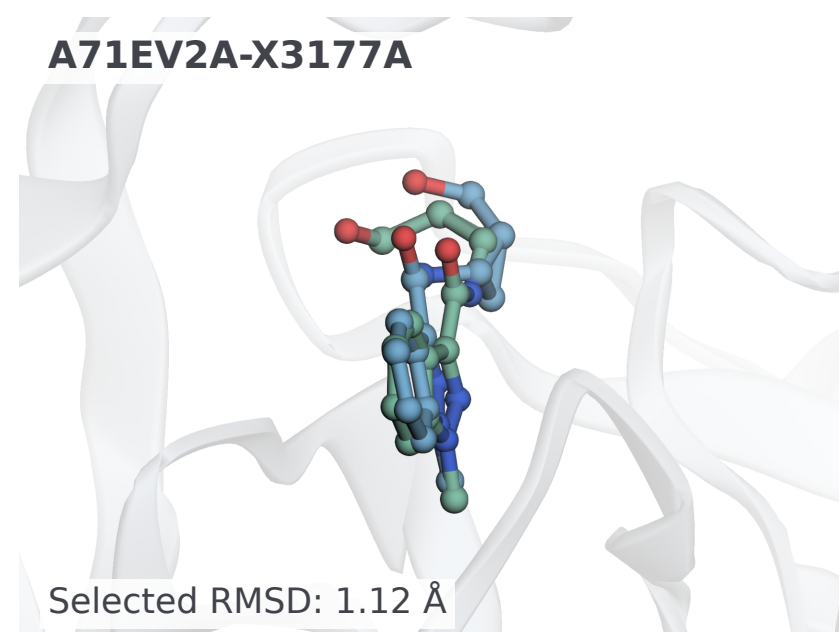
C PhiBench



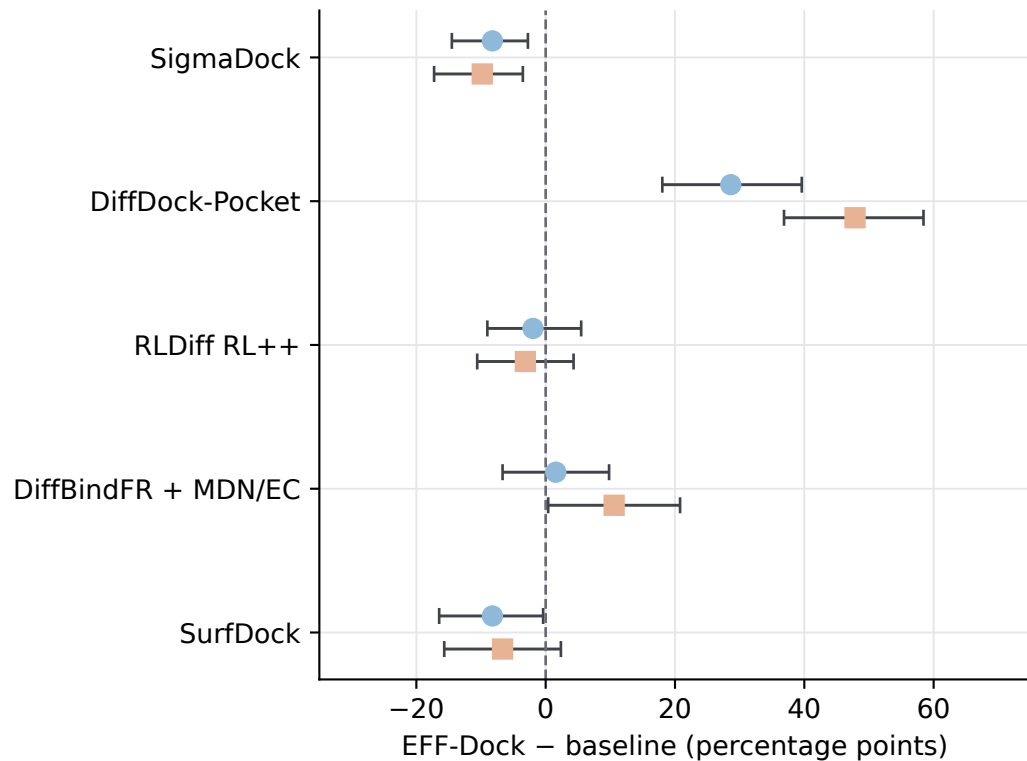
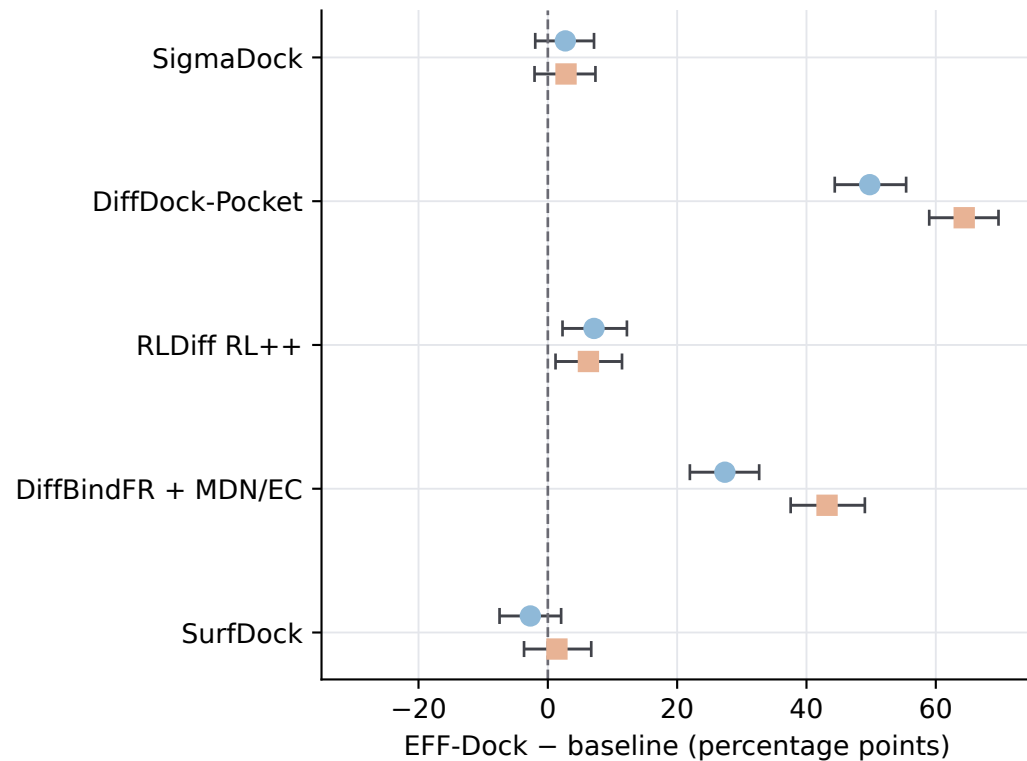
D FoldBench



E OpenBind

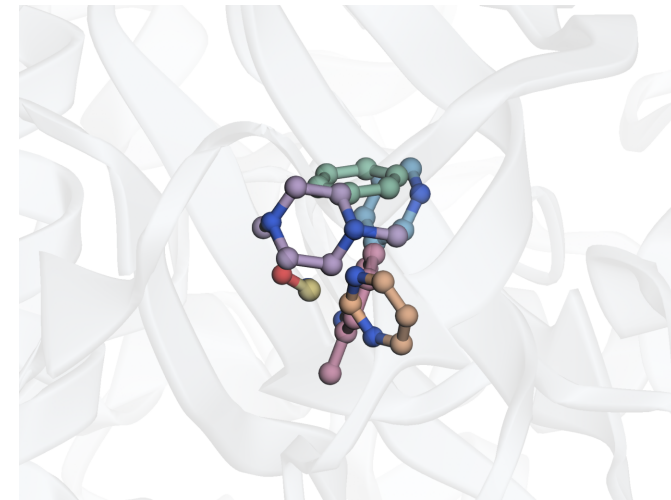


— Crystal carbon
 — Selected carbon
 — Raw / oracle carbon
 ● N
 ● O
 ● S
 ● P
 ● F
 ● Cl

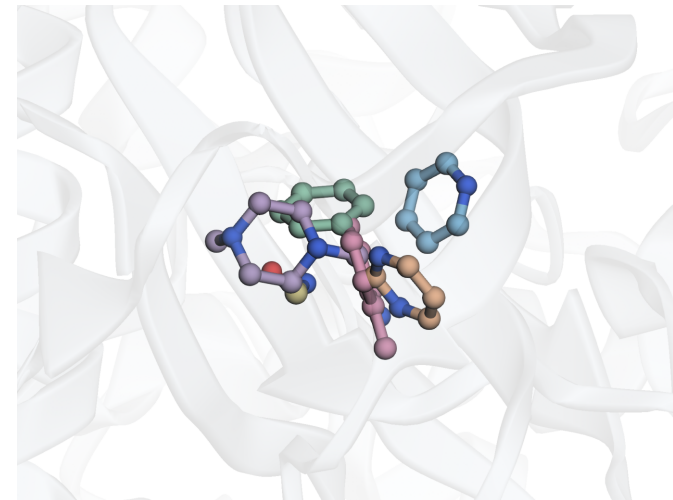
A Astex Diverse Set**B PoseBusters v2**

● RMSD success ■ PB-valid success

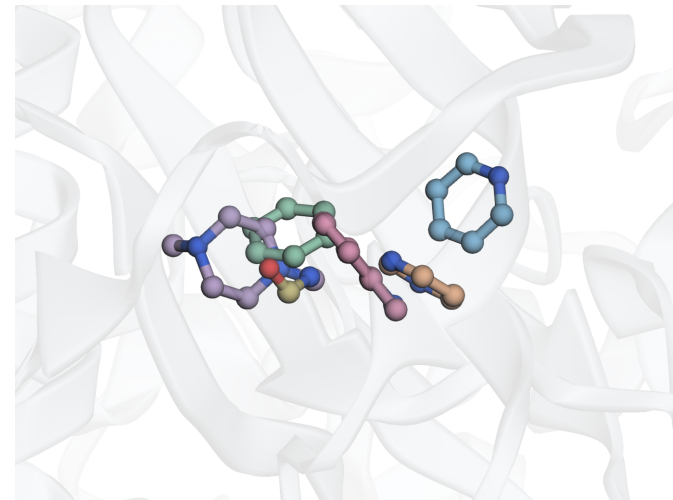
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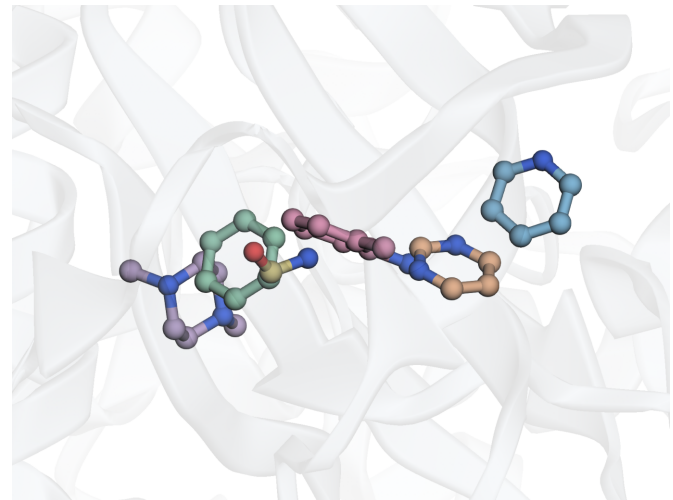
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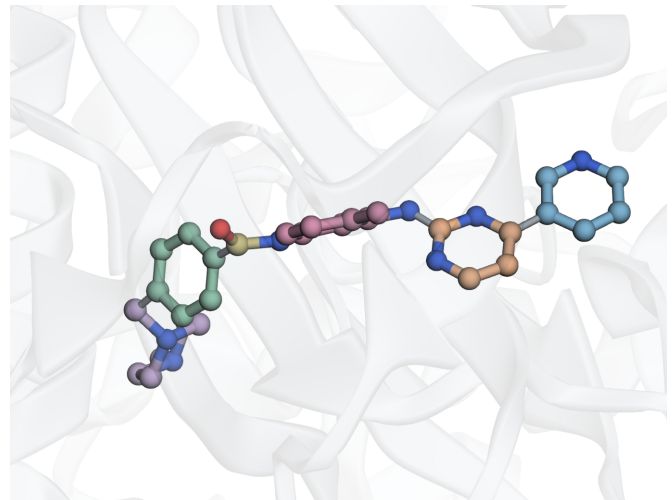
$t = 0.49$



$t = 0.78$



$t = 1.00$



— F1 — F2 — F3 — F4 — F5 — F6 ● N ● O